MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers Level 4 Tutor Guide

Holiday program for Level 4

Notice it. Test it. Explain why.

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Children whose school year covers Primary 4 topics (ages 10 to 11), going into P5, who enjoy thinking about maths. Every child gets the same pages and chooses how far to go
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Factor Hunters**	Factors and multiples	Making a list in order

Week 1b	2	**Big Number Puzzles**	Numbers to 100 000	Using a number line
Week 2a	3	**Operation Detectives**	Four operations	Order of operations
Week 2b	4	**Fraction Frontier**	Fractions	Drawing a fraction model
Week 3a	5	**Decimal Detectives**	Decimals	Place value to thousandths
Week 3b	6	**Angle Turns**	Angles	Estimating with 90° and 180°
Week 4a	7	**Area Architects II**	Area and Perimeter	Splitting and taking away
Week 4b	8	**Symmetry and Nets**	Symmetry and Nets	Visualising (folding in your head)
Week 5a	9	**Data Detectives**	Tables and line graphs	Reading scales
Week 5b	10	**Challenge Day**	Mixed heuristics	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens	Face to face	Online
0–4	**Sneak-peek puzzles**	Hello (0 to 1), then two sneak-peek puzzles on an idea we have not learnt yet (1 to 4): the first uses objects, a drawing or counting, the second is a thinking question. Guesses and “I do not know” are welcome. Nobody is marked and the answers do not decide who does what.	Own paper, held to the camera when the tutor says. Tutor looks only to plan help.	Children write on their own paper and hold it to the camera at the same moment, or show only the tutor on camera.
4–8	**Warm-up game and quick questions**	One short round of the game (4 to 6) and the quick questions (6 to 8). Everyone joins in.	Sit in a circle or horseshoe.	Cameras on. Answers by thumbs, hands or chat.
8–18	**Explore**	Hands-on in pairs. Objects first, no writing yet.	Materials box on each table.	Each child has their own materials at home. Pairs go to breakout rooms.
18–30	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes.	Whiteboard.	Shared slide and annotation tool.
30–34	**Brain break**	Stand, stretch, drink water.	A short movement game.	Stand up on camera. Count the stretches together.
34–80	**Worksheet**	Page 1 in pairs (34 to 50), check (50 to 53), short teaching (53 to 58), page 2 alone (58 to 76), check (76 to 80). Tutor asks “how do you know?” rather than giving answers.	Printed workbook, tutor circulates.	Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each.
80–85	**Talk time**	Share one answer and explain it. Compare methods.	Each child shares in turn.	Each child unmutes in turn.

85–88	**Exit ticket and stamp**	One question on a slip of paper. Stamp the inquiry journal. (Lesson 10: 85 to 87.)	Paper slip, tutor stamps.	Child writes the answer and shows the camera. Tutor stamps the next time or posts it.
88–90	**Looking back (2 minutes)**	Children fill in the small "Looking back" box on page 3, a page every child reaches. Write, tick or pass. Nobody is made to share. Never graded or collected. (Lesson 10: 87 to 89, certificates 89 to 90.)	Pencil and workbook.	Children write on the printed page. Passing is always allowed. A child who ticks "still stuck" gets a check-in in the open session.

Teaching habits that matter

- Ask "How do you know?" after every answer, right or wrong.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking ("You checked it two ways"), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child finishes early, use the Challenge or Extension question, not extra drill.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Cut-outs used: digit, point, sign and bracket cards (Lessons 2, 3, 5, 10), square grid (Lessons 7, 8), dot grid and cube net (Lesson 8), graph grids (Lesson 9), fraction wall (Lesson 4), right-angle testers (Lesson 6), and a paper clock, pizza circles and what-to-bring page (Lessons 4, 6). Children bring their own clear 180 degree protractor and a 30 cm ruler for Lesson 6.
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, scissors, a ruler, paper and a pencil.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child's page and hands.
- Hold up the answer page to the camera at the end of each lesson, or take a photo and send it through the approved channel.
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Factor Hunters	36 square tiles or squared paper per pair, mini whiteboards, coloured pencils, a 1 to 50 number strip.	Children cut 24 paper squares (or use squared paper) before the lesson and build rectangles on camera; the tutor marks the bus time line and the factor rainbow on the slides.
2	Big Number Puzzles	Digit cards 0 to 9 (one set per child: cut them from the Digit, point, sign and bracket cards page), mini whiteboards, a place-value chart (ten thousands to ones), a blank number line strip.	Children cut out the digit cards from the kit page before the lesson and move them on camera; the tutor marks the number lines on the slides.
3	Operation Detectives	Mini whiteboards and markers; number cards 0 to 9, sign cards +, −, ×, ÷ and two pairs of bracket cards (all on the Digit, point, sign and bracket cards page, one page per child); scrap paper for bar models.	Children write on mini whiteboards and hold them up to the camera. Before the lesson they cut out the cards page so they can move the number, sign and bracket cards on camera for the explore task.
4	Fraction Frontier	Paper strips (one per child, to fold into halves, quarters and eighths), the three pizza circles from the cut-out page (to cut into quarters), 30 counters, mini whiteboards and markers.	Before the lesson children cut out the 3 pizza circles from the kit page and cut each into 4 equal slices; they fold paper strips on camera and hold up mini whiteboards.
5	Decimal Detectives	Digit cards 0 to 9 and a decimal point card for each child (cut from the Digit, point, sign and bracket cards page), a place-value chart (ones, tenths, hundredths, thousandths), mini whiteboards, a metre ruler, play money.	Children cut out the digit cards and a decimal point from the kit page before the lesson and move them on camera. The tutor marks the zoom-in number lines on the slides.

6	Angle Turns	A protractor and a ruler for each child (bring from home: the kit page lists them), the paper clock from the cut-out page with a split pin (or a real clock), a square of paper for checking right angles, mini whiteboards, a compass card on the floor or wall.	Children use their own protractor and ruler and the paper clock cut out from the kit page. Self-view on screen may be mirrored: say turn towards your own right hand, not the screen's left. The 8-point compass is shown on the slides; children stand and turn on camera, with the screen as North.
7	Area Architects II	1 cm square tiles or 24 paper squares per child (cut the Square grid page into squares), 1 cm grid paper, scissors, a 30 cm ruler, mini whiteboards, a paper strip 8 squares by 2 squares for cutting.	Children cut 24 paper squares and some 1 cm grid strips from the Square grid page before the lesson and build on camera; the tutor draws on the grid figures on the slides.
8	Symmetry and Nets	Squared paper (2 cm squares), scissors, sticky tape, 6 paper squares per child (cut from the Square grid page), a dice, a small mirror (optional), mini whiteboards, coloured pencils.	Children cut out 6 paper squares from the Square grid page, and the cube net, before the lesson and fold on camera; the tutor clicks to reveal each fold on the slides and shades the grids.
9	Data Detectives	Mini whiteboards, a ruler and two coloured pencils per child, the blank graph grids from the kit page (or squared paper), the queue table for the explore task.	Children use the printed Graph grids page or squared paper and hold it up to the camera; the tutor plots on the slide graphs while children call out where each point goes.
10	Challenge Day	Four puzzle cards (A to D) for the explore stations (pairs choose three), mini whiteboards, counters or cubes, certificates, stamps.	Pairs go to breakout rooms with the puzzle cards (each pair chooses three of the four) and volunteers report back; the tutor ticks and crosses the logic grid on the slide as children call out.

Level 4 stories: the mathematicians

Each lesson has an optional 2-minute Story break at the brain break (after the mini-lesson, before the worksheet): one slide with a short true story and a Try it question, a Story corner box on the workbook page that every child reaches (the Investigation and Looking back page), and a line in the script. Skip it whenever time is short. Stories marked “accounts say” or with uncertain dates are labelled as such on purpose. Dates for people from long ago are estimates.

Lesson	Person	Country	Years	Link to the lesson
1	**Sophie Germain**	France (Paris)	1776-1831	Today you hunt for factors, and Sophie studied prime numbers, the numbers with only two factors.
2	**Leonardo of Pisa (Fibonacci)**	Italy (Pisa), schooled in Bugia, now Bejaia, Algeria	about 1170 - about 1240/1250	Today you read and build numbers up to 100 000, and place value is the idea that Leonardo's book helped spread.
3	**Katherine Johnson**	USA (West Virginia and Virginia)	1918-2020	Today you use add, subtract, multiply and divide, and Katherine's job was to get these calculations right and to check them.
4	**Mahavira**	India (Karnataka, possibly Mysore)	about 800 - about 870 (he worked around 850)	Today you compare and change fractions, and Mahavira's book explained how to work with fractions and split them into unit fractions.
5	**Simon Stevin**	Flanders (born in Bruges, now Belgium)	1548-1620	Today you work with tenths, hundredths and thousandths, which is exactly what Stevin's small book explained.
6	**Eratosthenes**	Cyrene (now Libya) and Alexandria, Egypt; ancient Greek world	about 276 BC - about 194 BC	Today you measure angles and turns, and Eratosthenes used the angle of a shadow to measure the whole Earth.

7	**Euclid**	Alexandria, Egypt (Greek-speaking world)	about 325 BC - about 265 BC (dates are estimates)	Today you cut and rearrange shapes and explain what happens to area, which is the idea in Euclid's rule for parallelograms.
8	**Emmy Noether**	Germany (born in Erlangen)	1882-1935	Today you explore symmetry, and Emmy showed that symmetry is useful as well as beautiful.
9	**Florence Nightingale**	Britain (born in Florence, Italy)	1820-1910	Today you read tables and line graphs, and Florence used tables and charts to make numbers easy to understand.
10	**Terence Tao**	Australia (born in Adelaide); now also works in the USA	born 1975	Today is Challenge Day, with competition-style puzzles like the ones in the Olympiad that Terence Tao entered three years in a row.

Safeguarding and tone: offer the story and the Try it question to the whole group, never to one child by name. Children may pass. No scores or tallies. Keep the Start here / Think harder / Go further names.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 8- and 9-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Lesson 1: Factor Hunters

Week 1 (first lesson) • Factors and multiples • Workbook: Lesson 1

Objective	Find all the factor pairs of a number by working in order, find common factors and common multiples to solve sharing and timetable puzzles, discover which numbers have an odd number of factors, and solve multiple-based logic riddles.
Thinking skills	Making a list in order, looking for a pattern, using clues one at a time, explaining why
Materials	36 square tiles or squared paper per pair, mini whiteboards, coloured pencils, a 1 to 50 number strip.
Online set-up	Children cut 24 paper squares (or use squared paper) before the lesson and build rectangles on camera; the tutor marks the bus time line and the factor rainbow on the slides.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome, Factor Hunters! Every number hides its factors inside it, and today we hunt them all down, without missing a single one."*

"We will also find out when two buses meet again, and why some numbers have an odd number of factors. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about factors and multiples. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and

the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Ali has 12 sweets. He puts them into bags so that every bag has the same number of sweets, with none left over. Use counters or draw dots. Write ONE way he can do it, like 2 bags of 6.	Any one of these is right: 1 bag of 12, 2 bags of 6, 3 bags of 4, 4 bags of 3, 6 bags of 2 or 12 bags of 1. The numbers of sweets in a bag (1, 2, 3, 4, 6, 12) are the factors of 12. A child with one way has had a win: you may ask the whole group, can anyone find another?
Sneak-peek 2 (thinking): Two lifts. Lift A stops at every 3rd floor (3, 6, 9 ...). Lift B stops at every 5th floor (5, 10, 15 ...). Which floors from 1 to 40 do BOTH lifts stop at?	Floors 15 and 30. They are in both lists: they are the multiples of 15.

DO A good reason sounds like: A factor goes INTO a number (the sweets in each bag). A multiple is in a times table (the lift floors). Write both lists and look for what they share.

4–8 min Warm-up game and quick questions

DO Fizz-Buzz. Count round the group from 1. Say "Fizz" for a multiple of 3, "Buzz" for a multiple of 4 and "Fizz-Buzz" for a multiple of both. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"We count round the group, one number each. If your number is in the 3 times table, say "Fizz". In the 4 times table, say "Buzz". In both, say "Fizz-Buzz"!"*

"If you slip, no problem: say "My go again" and pick someone to start the next round."

"Before we start: which will be the first Fizz-Buzz number? Write your guess on your whiteboard."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Is 7 a factor of 56?	Yes: 7 × 8 = 56
The first five multiples of 6?	6, 12, 18, 24, 30
Name a common multiple of 4 and 6.	12 (or 24, 36 ...)
Is 45 a multiple of 4?	No: 44 and 48 are
18 as a product, 3 ways?	1 × 18, 2 × 9, 3 × 6

8–18 min Explore (in pairs, with objects)

SAY *"You have 24 square tiles. You must use all of them to make one rectangle. How many different rectangles do you think you can make? Write your guess."*

DO 1. Take 24 square tiles.

2. Use ALL the tiles to make a rectangle. Write it as a multiplication, like 2 × 12.

3. Make as many different rectangles as you can. A 2 by 12 and a 12 by 2 count as the same.

4. How do you know you have found them all?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK start with one long row. How many tiles in the row? Now try 2 rows.

Not sure if 4 by 6 and 6 by 4 are different: turn your rectangle round. Is it the same shape?

Finished early: do it again with 36 tiles. Something special happens. What is it?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many rectangles did you make?	4: 1 × 24, 2 × 12, 3 × 8, 4 × 6.
How do you know you have them all?	Try 1, 2, 3, 4, 5, 6 rows in order. 5 rows does not work; 6 rows is the 4 × 6 turned round.
What do the side lengths tell us about 24?	They are the factors of 24: 1, 2, 3, 4, 6, 8, 12, 24.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Factor pairs of 36. I try 1, 2, 3 in order. 1 × 36. 2 × 18. 3 × 12. 4 × 9. Does 5 work? No, 36 is not in the 5 times table. 6 × 6. The next number, 9, is already in a pair, so the pairs have met. I stop. 36 has 9 factors."*

"12 apples and 18 oranges in identical bags, nothing left over. The number of bags must be a factor of 12 AND a factor of 18. The common factors are 1, 2, 3, 6. The most bags is 6, with 2 apples and 3 oranges in each."

"Buses every 6 minutes and every 8 minutes. I list the multiples: 6, 12, 18, 24 … and 8, 16, 24. The first number in both lists is 24, so they meet again after 24 minutes, not 48."

Worked example on the slides	Answer
All the factor pairs of 36. Try 1, 2, 3 … in order: 1 × 36, 2 × 18, 3 × 12, 4 × 9. 5 is not a factor. 6 × 6. The pairs have met. How many factors? Draw a factor rainbow.	9 factors: 1, 2, 3, 4, 6, 9, 12, 18, 36.

When do the buses meet again?. Bus A leaves every 6 minutes, bus B every 8 minutes. Both leave at 7:00 am. List the multiples of 6 and of 8. Which number is in both lists first?	24 minutes, so 7:24 am. 24 is the smallest common multiple of 6 and 8.

SAY *"The key idea today is: hunt in order. Factors: try 1, 2, 3 … in order until the pairs meet. Multiples: list them and look for the first number in both lists. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Listing factors at random and missing one, or stopping too early (for 48, forgetting 6 × 8), instead of trying 1, 2, 3 … in order until the pairs meet.
- Mixing up factors and multiples: saying 12 is a factor of 4, or listing 4, 8, 12 as the factors of 4.
- For common multiples, multiplying the two numbers every time (10 × 15 = 150 minutes) instead of listing the multiples to find the first one they share (30).

30–34 min Brain break

DO Factor freeze. The tutor calls a number. Stand up if your age, 10, is a factor of it. Sit down if it is not. Then try 3, 4 and 5.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Sophie Germain (France (Paris), 1776-1831). Today you hunt for factors, and Sophie studied prime numbers, the numbers with only two factors. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: 2 x 7 + 1 = 15. The factors of 15 are 1, 3, 5 and 15, so 15 is not prime and 7 is not a Sophie Germain prime. For 5: 2 x 5 + 1 = 11, whose only factors are 1 and 11, so 11 is prime and 5 is a Sophie Germain prime.

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 1 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “How do you know you have found every factor pair? Where did you stop?”
- Ask: “Is that a factor or a multiple? Which is bigger?”
- Ask: “Which list did you write: the factors, or the multiples? Which do you need for this question?”

IF STUCK Hint 1: “Try 1, 2, 3, 4 … in order. Does it go in exactly?”

Hint 2: “Draw the rectangles, or make them with tiles.”

Hint 3: “Write the first few multiples of each number in two rows. Circle the numbers in both rows.”

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: How many factors does 72 have? List them in order.	12 factors: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.
Workbook Think harder 2: One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?	18 seconds (6, 12, 18 and 9, 18).
Workbook Think harder 3: Find all the common factors of 24 and 36.	1, 2, 3, 4, 6, 12
Workbook Think harder 4: Find ALL the numbers from 1 to 60 that have exactly 3 factors. What do the numbers have in common?	4, 9, 25 and 49. Each is a number times itself (2 × 2, 3 × 3, 5 × 5, 7 × 7), where that number has only 2 factors. 16 = 4 × 4 has 5 factors and 36 has 9.
Workbook Go further 1: Bus A leaves a stop every 6 minutes and Bus B every 9 minutes. Bus C leaves every ___ minutes. All three leave together at 9:00 am, and they next leave together at 9:36 am. What could the gap for Bus C be? Find ALL the whole-minute answers. Then make up a puzzle like this one for a partner.	4, 12 or 36 minutes. The smallest common multiple of 6 and 9 is 18, so Bus C's gap must be a factor of 36 that makes the first meeting 36 (not 18). 1, 2, 3, 6, 9 and 18 all give 18, so they do not work. Puzzles will vary: check that the partner's answer really gives the stated time.
Workbook Go further 2: I am a number less than 100. When I am divided by 2, 3, 4, 5 or 6, there is always 1 left over. I am bigger than 1. Who am I?	61. One less than the number is a common multiple of 2, 3, 4, 5 and 6: 60. 60 + 1 = 61 (the next one, 121, is too big).

Workbook Go further 3: There are 20 closed lockers, numbered 1 to 20. Child 1 opens every locker. Child 2 changes every 2nd locker (opens it if it is closed, closes it if it is open). Child 3 changes every 3rd locker, and so on, up to child 20. Which lockers are open at the end?	1, 4, 9 and 16. Each locker is changed once for every factor of its number. Only numbers with an odd number of factors (1, 4, 9, 16: a number times itself) end up open.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Ali says: bigger numbers always have more factors. Who agrees? Can you find an example that proves him wrong?"*

"Why did the 36-tile rectangles give an odd number of factors?"

Discussion question	Listen for
Ali says: "Bigger numbers always have more factors." Is he right? Give an example to show it.	Ali is wrong. 24 has 8 factors but 29, which is bigger, has only 2 (1 and 29). 36 has an odd number of factors because one of its pairs is 6 × 6, and 6 is only counted once.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Write all the factors of 28. Then find the smallest number that is a multiple of both 6 and 9.	1, 2, 4, 7, 14, 28. The smallest common multiple of 6 and 9 is 18.

SAY *"Today you hunted factors in order, so none could hide. You found common factors for sharing and common multiples for timetables."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I wrote the factor pairs in order. The hardest part was knowing when to stop listing factors. What helped me: writing the pairs in order and asking my partner." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Which numbers from 1 to 50 have exactly 4 factors? Is there a pattern? (There are 15: 6, 8, 10, 14, 15, 21, 22, 26, 27, 33, 34, 35, 38, 39, 46.)

Answer key (workbook)

Q1 (a) 1 × 48, 2 × 24, 3 × 16, 4 × 12, 6 × 8 (b) 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors (c) 7 is not a factor, and the next number, 8, is already in a pair (6 × 8). The pairs have met, so there are no more.

Q2 6 ways: 2 rows of 20, 4 rows of 10, 5 rows of 8, 8 rows of 5, 10 rows of 4, 20 rows of 2.

Q3 (a) 18: 1, 2, 3, 6, 9, 18. 24: 1, 2, 3, 4, 6, 8, 12, 24 (b) 1, 2, 3, 6 (c) 6 packs, each with 3 red and 4 blue beads.

Q4 (a) Bus 33: 6:00, 6:10, 6:20, 6:30 … Bus 51: 6:00, 6:15, 6:30 … Together again at 6:30 am. (b) Every 30 minutes: 6:00, 6:30, 7:00, 7:30, 8:00, so 5 times.

Q5 (a) 6: 4 factors, 9: 3, 10: 4, 15: 4, 16: 5, 25: 3 (b) 9 and 25. Each is a number times itself (3 × 3 and 5 × 5). (c) Yes: 1, 7, 49. It is 7 × 7, a number times itself, like 9 and 25.

Q6 (a) 36 (multiples of 12 between 30 and 70 are 36, 48, 60; 48 is a multiple of 8; in 60 the ones digit is smaller). (b) Ravi. 6 = 2 × 3, so every multiple of 6 is in the 3 times table. Mei is wrong: 9 is a multiple of 3 but not of 6.

Investigation Bus meeting times (What if we change …? What stays the same? Find all the ways.) Easy first step: Write the first five times Bus A leaves (7:00, 7:06, 7:12, …) and the first five for Bus B (7:00, 7:08, …). Circle a time that is in both lists. Model answer: (a) 18, 12, 30, 42 minutes. (b) The meeting time is always a multiple of both gaps (the smallest common multiple), and never smaller than the bigger gap. When the gaps share no factor except 1 (6 and 5, 6 and 7) the meeting time is their product (30, 42). Accept any true pattern. (c) Six pairs: 2 and 12, 3 and 4, 3 and 12, 4 and 6, 4 and 12, 6 and 12 (checked by brute force). Look for a systematic list (fix one gap, test the others) rather than guesses. A child with 3 or 4 pairs who can explain how they know is doing well. Extension: choose your own meeting time (such as 30) and find all the pairs.

Go further What if we use 36 tiles? What is special about one of the rectangles? 5 rectangles: 1 × 36, 2 × 18, 3 × 12, 4 × 9 and 6 × 6. The 6 × 6 is a square: 6 is its own partner, so 36 has an odd number (9) of factors.

Story corner try-it Sophie Germain: Is 2 × 7 + 1 a prime? List the factors of your answer. Then try 5. Answer: 2 x 7 + 1 = 15. The factors of 15 are 1, 3, 5 and 15, so 15 is not prime and 7 is not a Sophie

Germain prime. For 5: 2 x 5 + 1 = 11, whose only factors are 1 and 11, so 11 is prime and 5 is a Sophie Germain prime.

Sneak-peek puzzles (1) Any one of these is right: 1 bag of 12, 2 bags of 6, 3 bags of 4, 4 bags of 3, 6 bags of 2 or 12 bags of 1. The numbers of sweets in a bag (1, 2, 3, 4, 6, 12) are the factors of 12. A child with one way has had a win: you may ask the whole group, can anyone find another? (2) Floors 15 and 30. They are in both lists: they are the multiples of 15.

Lesson 2: Big Number Puzzles

Week 1 (second lesson) • Numbers to 100 000 • Workbook: Lesson 2

Objective	Round numbers to the nearest 10, 100 and 1000 using a number line, work backwards from a rounded number to find the smallest and largest possible numbers, use rounding to estimate and check, and solve 5-digit number riddles, digit-card puzzles and number patterns.
Thinking skills	Using a number line, working backwards, making a list in order, using the clues one at a time, looking for a pattern
Materials	Digit cards 0 to 9 (one set per child: cut them from the Digit, point, sign and bracket cards page), mini whiteboards, a place-value chart (ten thousands to ones), a blank number line strip.
Online set-up	Children cut out the digit cards from the kit page before the lesson and move them on camera; the tutor marks the number lines on the slides.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome back, inquirers! Today our numbers get BIG: all the way up to 100 000. That is more than the number of people who fit in the National Stadium."*

"Big numbers are often rounded. Today we find out what a rounded number can hide, and we solve riddles about 5-digit numbers. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about rounding. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Draw a number line from 7 000 to 8 000. Put a dot at 7 500, halfway. Then put a dot at 7 200. Is the 7 200 dot nearer to 7 000 or to 8 000?	Nearer to 7 000. 7 200 is only 200 from 7 000 but 800 from 8 000, and it is left of the halfway dot 7 500.
Sneak-peek 2 (thinking): A shop says "about 8 000 people came" (rounded to the nearest thousand). Which of these could be the real number: 7 499, 7 500, 8 499, 8 500? Hint: draw a number line from 7 000 to 9 000.	7 500 and 8 499. The real number is from 7 500 to 8 499 (7 499 rounds down to 7 000; 8 500 rounds up to 9 000).

DO A good reason sounds like: Find the two round numbers and the halfway mark. Halfway or more rounds up.

4–8 min Warm-up game and quick questions

DO Place It Right (5 digits). The tutor turns over digit cards one at a time. Write each in a box straight away. Round 1: biggest wins. Round 2: closest to 50 000 wins. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"Draw five boxes on your whiteboard: ten thousands, thousands, hundreds, tens, ones. I turn over one digit card at a time. Write it in a box straight away. No moving it later!"*

"Round 1: make the biggest number. Round 2: make the number closest to 50 000. Where will you put a 4? A 5?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is the 7 worth in 47 352?	7000
Sixty thousand and forty-five in numerals?	60 045
1000 more than 59 600?	60 600
40 000 + 3000 + 50 + 8 = ?	43 058
36 815 to the nearest 1000?	37 000

8–18 min Explore (in pairs, with objects)

SAY *"You have the cards 2, 4, 5, 7 and 9. How many of the numbers you can make will round to 50 000 to the nearest thousand? Write your guess."*

DO 1. Take the digit cards 2, 4, 5, 7 and 9.

2. Make a 5-digit number using each card once. Write it down.

3. Round it to the nearest thousand. Does it round to 50 000?

4. Find ALL the numbers that round to 50 000. Which is closest to 50 000?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK which ten thousands digit could give a number near 50 000? There is no 5 0 … because there is no 0 card.

Not sure about one number: draw a number line from 49 000 to 50 000. Where is the halfway mark?

Finished early: which number you can make is closest to 70 000?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which numbers round to 50 000?	49 527, 49 572, 49 725, 49 752.
How do you know you have them all?	They must be from 49 500 to 50 499. With no 0 card, they start 49 and then 5 or 7: 2 numbers each.
Which is closest to 50 000?	49 752 (248 away).

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"36 482 to the nearest thousand. It is between 36 000 and 37 000. Halfway is 36 500. 36 482 is just below halfway, so it is nearer 36 000."*

"A number rounds to 25 000. The smallest it could be is 24 500, because 24 500 is exactly halfway and halfway rounds up. The largest is 25 499. 25 500 would round up to 26 000."

"7 × 3 089. I round 3 089 to 3 000, so the estimate is 21 000. I rounded DOWN, so the real answer must be a bit more: 21 623."

Worked example on the slides	Answer
Round 36 482 to the nearest thousand. Which two thousands is it between? 36 000 and 37 000. Halfway is 36 500. Is 36 482 before or after halfway?	36 000 (it is before halfway). To the nearest hundred it is 36 500.

Work backwards: what rounds to 25 000?. Rounded to the nearest thousand, a number is 25 000. The halfway marks are 24 500 and 25 500. What are the smallest and largest whole numbers it can be?	Smallest 24 500, largest 25 499 (25 500 rounds up to 26 000).

SAY *"The key idea today is: find the halfway mark. To round, find the two round numbers and the halfway mark. Halfway or more rounds up. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Rounding by looking at the wrong digit, or forgetting to carry when the digit to change is a 9 (63 950 to the nearest hundred written as 63 900 or 63 000 instead of 64 000).
- Working backwards with the wrong limits: saying the largest number that rounds to 63 000 is 63 500 (it rounds up to 64 000) instead of 63 499.
- Digit-card puzzles: putting 0 in the ten thousands place, or making the biggest number first and then hoping it is even, instead of choosing the ones digit first.

30–34 min Brain break

DO Human number line. Five children hold 10 000, 20 000, 30 000, 40 000, 50 000. The tutor calls a number: everyone points to the nearest ten thousand.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Leonardo of Pisa (Fibonacci) (Italy (Pisa), schooled in Bugia, now Bejaia, Algeria, about 1170 - about 1240/1250). Today you read and build numbers up to 100 000, and place value is the idea that Leonardo's book helped spread. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: Biggest: 97 420. Smallest: 20 479 (the 0 cannot go first, or it would not be a 5-digit number).

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 2 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “Which two round numbers is it between? Where is the halfway mark?”
- Ask: “Could the number be bigger than that and still round to the same number?”
- Ask: “Which clue gives you the fewest numbers to try? Start there.”

IF STUCK Hint 1: “Draw a number line. Mark the two round numbers and halfway.”

Hint 2: “Look only at the digit to the right of the place you round to.”

Hint 3: “For odd or even, choose the ones digit FIRST.”

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: I am a 5-digit number. All my digits are the same. Rounded to the nearest ten thousand, I am 60 000. Who am I?	55 555 (66 666 rounds to 70 000; 44 444 rounds to 40 000).
Workbook Think harder 2: Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.	69 741 (259 away; the closest number above 70 000 is 71 469, which is 1469 away).
Workbook Think harder 3: Find the rule and write the next two numbers: 17 250, 19 500, 21 750, ____, ____	Add 2250 each time: 24 000, 26 250.
Workbook Think harder 4: How many whole numbers round to 3500 when rounded to the nearest hundred?	100 (from 3450 to 3549).
Workbook Go further 1: How many 5-digit numbers have digits that add up to 3?	15. Ten thousands digit 1: the other 2 goes as 2 in one place (4 ways) or 1 and 1 in two places (6 ways): 10. Ten thousands digit 2: 4 ways. Ten thousands digit 3: 1 way (30 000). 10 + 4 + 1 = 15.
Workbook Go further 2: Make up your own number riddle about a 5-digit number. Use a rounding clue, a clue about the digits and an odd-or-even clue, so that exactly ONE number fits. Check your riddle: list every number that fits the rounding clue, then cross out the ones that break the other clues. Swap riddles with a partner.	Riddles will vary. Check that the rounding clue gives the right range and that only one number is left. Model riddle: "Rounded to the nearest thousand I am 30 000. All my digits are different. My digits add up to 12. I am odd." The only number is 30 261.

Workbook Go further 3: The distance counter on Mr Tan's car shows 39 993 km. This number is a palindrome: it reads the same forwards and backwards. What is the next palindrome the counter will show? How many more kilometres is that?	40 004, which is 11 km more.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Siti says 49 999 rounds to 50 000 to the nearest 10, 100 AND 1000. Is she right?"*

"Ravi says 45 000 rounds to 40 000 to the nearest ten thousand. Is he right?"

Discussion question	Listen for
Siti says 49 999 rounds to 50 000 to the nearest 10, the nearest 100 and the nearest 1000. Ravi says 45 000 rounds to 40 000 to the nearest ten thousand. Who is right?	Siti is right: 49 999 is only 1 away from 50 000, so it rounds to 50 000 every time. Ravi is wrong: 45 000 is exactly halfway between 40 000 and 50 000, and halfway rounds up, so it rounds to 50 000.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Round 74 652 to the nearest thousand. Then write the smallest whole number that rounds to 74 000 to the nearest thousand.	75 000. The smallest is 73 500.

SAY *"Today you used number lines to round, and you worked backwards to find what a rounded number could really be."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes,*

no talking."

"Model one yourself first: "Something I did well: I drew the number line. The hardest part was working out which way a number like 450 rounds. What helped me: marking the halfway point on the number line." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

How many whole numbers round to 45 000 to the nearest thousand AND to 50 000 to the nearest ten thousand? (500: from 45 000 to 45 499.)

Answer key (workbook)

Q1 47 285: 47 290, 47 300, 47 000. 63 950: 63 950, 64 000, 64 000. Try next: 9049: 9050, 9000, 9000. 80 503: 80 500, 80 500, 81 000.

Q2 (a) Smallest 62 500, largest 63 499 (b) 5345, 5346, 5347, 5348, 5349, 5350, 5351, 5352, 5353, 5354 (10 numbers)

Q3 (a) 6 × 5 000 = 30 000 (b) Less: 4 982 was rounded up to 5 000, so each day was counted as a little more than it really is. (c) 29 892

Q4 (a) 47 621 (b) 10 001, 10 010, 10 100, 11 000, 20 000 (5 numbers)

Q5 (a) 26 450, 27 450 (b) 76 800, 75 300 (c) 99 990, 100 000 (d) 32 500

Q6 (a) 86 530 (b) 30 685 (c) 60 358

Investigation Double rounding (Is it always true? Test and explain.) Easy first step: Round 3 280 to the nearest hundred. Then round your answer to the nearest thousand. Now round 3 280 straight to the nearest thousand. Did you get the same answer both ways? Model answer: Mei is not always right. 3 280 gives 3 300 then 3 000, and straight to 3 000: the same. 1 449 gives 1 400 then 1 000, and straight to 1 000: the same. 1 450 gives 1 500 then 2 000, but straight to the nearest thousand it is 1 000: different. Every number whose last three digits are 450 to 499 breaks the idea (checked by brute force for every whole number below 100 000). Why: rounding to the nearest hundred first turns a number just BELOW the halfway mark into exactly the halfway mark, which then rounds up. Look for children who test numbers near the halfway mark instead of only easy numbers, and who say "I tested ..." before they say "I think ...".

Go further Which number you can make is closest to 70 000? Why can you not make 70 000? 72 459 (2 459 away). 70 000 needs a 0 card and there is none; with the cards 2, 4, 5, 7 and 9 the closest below is 59 742 (10 258 away).

Story corner try-it Leonardo of Pisa (Fibonacci): Use the digits 7, 0, 4, 9 and 2 once each. Write the biggest and the smallest 5-digit number you can. Answer: Biggest: 97 420. Smallest: 20 479 (the 0

cannot go first, or it would not be a 5-digit number).

Sneak-peek puzzles (1) Nearer to 7 000. 7 200 is only 200 from 7 000 but 800 from 8 000, and it is left of the halfway dot 7 500. (2) 7 500 and 8 499. The real number is from 7 500 to 8 499 (7 499 rounds down to 7 000; 8 500 rounds up to 9 000).

Lesson 3: Operation Detectives

Week 2 (first lesson) • Four operations • Workbook: Lesson 3

Objective	Use the order of operations (brackets first, then multiplication and division from left to right, then addition and subtraction from left to right), find missing signs and brackets to hit a target, work backwards through a chain of operations, and solve "times as many" problems with a bar model.
Thinking skills	Order of operations, estimating then testing, guess and check, working backwards, drawing a bar model, explaining why a number trick works
Materials	Mini whiteboards and markers; number cards 0 to 9, sign cards +, −, ×, ÷ and two pairs of bracket cards (all on the Digit, point, sign and bracket cards page, one page per child); scrap paper for bar models.
Online set-up	Children write on mini whiteboards and hold them up to the camera. Before the lesson they cut out the cards page so they can move the number, sign and bracket cards on camera for the explore task.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome back, detectives! Today the clues are the signs: +, −, × and ÷, and the brackets that boss them around."*

"The same numbers can give very different answers. Our job is to find out why, and to make numbers do what we want. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about order of operations. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may

choose: carry on, or try Think harder. When you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Ali has 2 bags with 4 sweets in each bag, and 3 loose sweets. Draw it. How many sweets has he got? Write a number sentence that matches your drawing.	11 sweets: 2 × 4 + 3 = 8 + 3 = 11. The bags are counted first (×), then the loose sweets are added (+).
Sneak-peek 2 (thinking): Put ONE pair of brackets in 2 + 3 × 4 so that the answer is 20.	(2 + 3) × 4 = 5 × 4 = 20. With no brackets the answer is 14, because 3 × 4 is done first.

DO A good reason sounds like: A story helps: the bags are counted first, then the loose ones are added. Brackets change which part is done first.

4–8 min Warm-up game and quick questions

DO Target Numbers. Use the cards 2, 4, 5 and 9 (each once at most) with +, −, × and ÷ to hit 13, 22, 36 and 81. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"Here are four cards: 2, 4, 5 and 9. You can use each card once, with plus, minus, times or divide."*

"When a target appears, race to hit it on your whiteboard. Then look for a second way!"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
25 × 4 = ?	100
4 800 ÷ 6 = ?	800
99 × 6 = ?	594
1 000 − 368 = ?	632
45 × 20 = ?	900

8–18 min Explore (in pairs, with objects)

SAY *"Look at 6 + 4 × 3 − 2. If you can put ONE pair of brackets anywhere, how many different answers do you think you can make? Write your guess in the corner of your whiteboard."*

DO 1. Write 6 + 4 × 3 − 2 on your whiteboard and work it out.

2. Put ONE pair of brackets somewhere. Work it out again.

3. Move the brackets to other places. Write every answer.

4. How many different answers? Which is the biggest? The smallest?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK put the brackets round 6 + 4 first. What do you work out first now?

Not sure if you have them all: "Make a list of every place the brackets can go: round two numbers, round three numbers."

Finished early: can TWO pairs of brackets make a new answer?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many different answers did you find?	3 answers: 16, 28 and 10.
Where do the brackets go for the biggest answer?	(6 + 4) × 3 − 2 = 28.
Why do some brackets not change the answer?	Brackets round 4 × 3 change nothing: × is done first anyway.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"40 − 8 × 3 + 6 ÷ 2. No brackets, so I look for × and ÷ first: 8 × 3 is 24 and 6 ÷ 2 is 3. Now I have 40 − 24 + 3. Only + and − are left, so I go from left to right: 16, then 19."*

"8, 5 and 3 must make 37. Even adding all three only makes 16, so there must be a times. 8 × 5 is 40, and 40 − 3 is 37. Let me check by doing it in the right order: 8 × 5 − 3 = 37."

"21 came out of the machine. The last step was divide by 2, so I undo it: times 2 gives 42. Then undo take away 6: add 6, 48. Then undo times 4: divide by 4, 12. I check by going forwards."

Worked example on the slides	Answer
Order of operations. 40 − 8 × 3 + 6 ÷ 2 × and ÷ first: 24 and 3. 40 − 24 + 3, left to right.	19
Missing signs. 8 ? 5 ? 3 = 37 (each ? is a sign) 37 is big, so try ×: 8 × 5 = 40. 40 and 3 make 37 how?	8 × 5 − 3 = 37

SAY *"The key idea today is: order matters. Brackets first. Then × and ÷, then + and −, each from left to right. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The “your turn” slides are optional: skip them if you are short of time. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Working strictly from left to right and ignoring the order: 40 − 8 × 3 done as 32 × 3.
- Thinking addition always comes before subtraction (or multiplication before division) instead of working left to right: 20 − 8 + 2 answered 10, 72 ÷ 8 × 3 answered 3.
- Working backwards but using the same operation instead of the opposite one (for − 18, taking away 18 again instead of adding 18).

30–34 min Brain break

DO Odd or even stand-up. I read a number sentence: stand up if the answer is even, crouch if it is odd. 3 + 4 × 2, then (3 + 4) × 2, then 20 ÷ 4 + 1, then 20 ÷ (4 + 1).

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today’s mathematician is Katherine Johnson (USA (West Virginia and Virginia), 1918-2020). Today you use add, subtract, multiply and divide, and Katherine's job was to get these calculations right and to check them. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked. The story says “accounts say” on purpose: do not turn it into a certain fact.

To skip: say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on the story. Try it answer: 6 x 50 = 300, and 6 x 2 = 12, so 6 x 48 = 300 - 12 = 288. It checks out. You can also check with division: 288 divided by 6 = 48.

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 3 aloud. Say “Pairs first. Talk before you write.”

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: "Which part do you work out first? Point to it and underline it."
- Ask: "Is your target big or small compared with the numbers? What does that tell you?"
- Ask: "What is the opposite of this step? How do you undo it?"

IF STUCK Hint 1: "Underline the part in brackets, then the × and ÷ parts. Do those first."

Hint 2: "Estimate first. A big target usually needs a ×. Try one sign, then test it."

Hint 3: "Working backwards, start at the END and undo each step: × becomes ÷, − becomes +."

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask "How did you decide?" and "Is there another way?" rather than marking it right or wrong. The deck has a slide marked ON DEMAND (and in Lesson 5 two): it is not part of the five-minute teaching before page 2. Show it only while children work on page 2, when a child chooses that choice question, and make the offer to the whole group.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: There are 2 340 red and blue beads. There are 4 times as many blue beads as red beads. How many red beads are there?	468 (5 units = 2 340, 1 unit = 468).
Workbook Think harder 2: Put ONE pair of brackets in 3 + 5 × 6 − 2 to make the answer 46.	(3 + 5) × 6 − 2 = 8 × 6 − 2 = 46.
Workbook Think harder 3: Write +, −, × or ÷ in each gap to make this true: 12 __ 4 __ 5 = 15	12 ÷ 4 × 5 = 15 (the only way).
Workbook Think harder 4: Think of a number. Multiply it by 5. Add 35. Divide by 5. Take away the number you started with. The answer is always 7. Explain why it always works. A bar model can help.	Multiplying by 5 and then dividing by 5 gets back to your number. The 35 divided by 5 adds 7. Taking away your number leaves the 7. So the answer is always 7.
Workbook Go further 1: Put + or − in each gap so that the answer is 7: 1 __ 2 __ 3 __ 4 __ 5 __ 6 = 7. Find ALL the ways.	2 ways: 1 − 2 + 3 + 4 − 5 + 6 = 7 and 1 + 2 − 3 − 4 + 5 + 6 = 7. (All + makes 21, which is 14 too many. Changing a + to a − takes that number away twice, so the numbers we change must add up to 7: 2 and 5, or 3 and 4.)
Workbook Go further 2: Mei spent half of her money on a book. Then she spent $6 on lunch. Then she spent half of what was left on a gift. She had $9 left. How much money did she have at first?	$48. Work backwards: $9 was half of what was left, so before the gift she had $18. Before lunch: $18 + $6 = $24. That was half, so at first she had $48.
Workbook Go further 3: The number trick: think of a number, multiply by 5, add 35, divide by 5, take away your number. It always ends with 7. (a) What must you change so that it always ends with 10? (b) Make up your own number trick with four steps that always ends with 3. Test it with two numbers and explain why it works.	(a) Change the +35 to +50: the added number divided by 5 is what is left at the end. (b) Answers vary. Model: × 4, + 12, ÷ 4, − your number: always 3, because × 4 then ÷ 4 gets the number back, and 12 ÷ 4 = 3 is added.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Ben says 20 − 8 + 2 = 10 because you add first. Mei says 14. Who is right?"*

"How did you know which sign to try first in the missing-sign puzzles?"

Discussion question	Listen for
Ben says 20 − 8 + 2 = 10 because you add first. Mei says the answer is 14. Who is right, and why?	Mei is right. When there are only + and − (or only × and ÷), we work from left to right: 20 − 8 = 12, then 12 + 2 = 14. Ben worked out 8 + 2 first, which would need brackets: 20 − (8 + 2) = 10.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Work out 50 − 6 × (3 + 4).	8 (brackets first: 7; then 6 × 7 = 42; then 50 − 42 = 8).

SAY *"Today you made the signs and brackets work for you, and you worked backwards like real detectives."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I worked out the order step by step. The hardest part was remembering to do × before −. What helped me: underlining the part to do first." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout

room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Use four 4s and any of +, −, ×, ÷ and brackets to make every number from 0 to 9. (For example 0 = 4 + 4 − 4 − 4, 1 = (4 + 4) ÷ (4 + 4), 2 = 4 ÷ 4 + 4 ÷ 4, 3 = (4 + 4 + 4) ÷ 4, 4 = (4 − 4) × 4 + 4, 5 = (4 × 4 + 4) ÷ 4, 6 = (4 + 4) ÷ 4 + 4, 7 = 4 + 4 − 4 ÷ 4, 8 = 4 + 4 + 4 − 4, 9 = 4 + 4 + 4 ÷ 4.)

Answer key (workbook)

Q1 (a) 106 (b) 250 (c) 27 (d) 98 (e) 12

Q2 (a) 7 × 6 − 4 = 38 (b) 48 ÷ 6 × 2 = 16 (c) 9 ÷ 3 + 5 = 8 (d) 12 ÷ 3 + 2 × 5 = 14. Each puzzle has only one answer.

Q3 (a) 41 (b) (4 + 8) × 5 − 3 = 57 (c) 4 + 8 × (5 − 3) = 20

Q4 (a) 23. Work backwards: 30 × 4 = 120, 120 + 18 = 138, 138 ÷ 6 = 23. (b) Check: 23 × 6 = 138, 138 − 18 = 120, 120 ÷ 4 = 30.

Q5 The answer is always 7 (for example 3: 15, 50, 10, 7). (a) It is always 7. (b) Blanks: number, 7, 7. Multiplying by 5 and then dividing by 5 gets you back to your number. The 35 divided by 5 adds 7. Taking away your number leaves 7.

Q6 (a) Chicken rice: 3 units. Laksa: 1 unit. 4 units = 124. (b) 1 unit = 124 ÷ 4 = 31. Chicken rice = 3 × 31 = 93 plates. (c) 2 units = 62 more plates.

Q7 (a) 5, 8, 13, 21, 34 (21 − 13 = 8, 13 − 8 = 5, 13 + 21 = 34). (b) 4, 7, 11, 18, 29.

Investigation Make the target (Find all the ways. Make your own and swap.) Easy first step: Use only + and −. With 2, 3, 4 and 5 (each once) can you make 6? 8? 10? Write a number sentence for each. Model answer: Every target from 1 to 12 can be made (all of 1 to 20 can, checked by brute force). Model sentences: 1 = (5 − 4) × (3 − 2); 2 = 5 − 4 + 3 − 2; 3 = 5 × 2 − 4 − 3; 4 = 2 + 3 + 4 − 5; 5 = (2 + 3 − 4) × 5; 6 = 2 + 3 − 4 + 5; 7 = 2 + (4 − 3) × 5; 8 = 5 + 4 + 2 − 3; 9 = 2 + 3 × 4 − 5; 10 = (2 × 3 − 4) × 5; 11 = 2 × 5 − 3 + 4; 12 = 2 × (3 + 5) − 4. Many other correct sentences exist: accept any that work out. Look for children who work each sentence out in the right order and check it, and who use brackets on purpose.

Go further What if you may use TWO pairs of brackets? Can you get an answer that one pair could not? No. Two pairs still give only 16, 28 and 10 (every placement checked by computer). Brackets only change what is done first, and here there are only three different orders that matter: the × first (16), (6 + 4) first (28) or (3 − 2) first (10).

Story corner try-it Katherine Johnson: Check that 6 × 48 = 288 using a different method from your first one. Hint: 48 is close to 50. Answer: 6 x 50 = 300, and 6 x 2 = 12, so 6 x 48 = 300 - 12 = 288. It checks out. You can also check with division: 288 divided by 6 = 48.

Sneak-peek puzzles (1) 11 sweets: 2 × 4 + 3 = 8 + 3 = 11. The bags are counted first (×), then the loose sweets are added (+). (2) (2 + 3) × 4 = 5 × 4 = 20. With no brackets the answer is 14, because 3 × 4 is done first.

Lesson 4: Fraction Frontier

Week 2 (second lesson) • Fractions • Workbook: Lesson 4

Objective	Change between improper fractions and mixed numbers using fraction bars and number lines, find a fraction of a set (and the whole set from a part) with a bar model, and compare fractions with different denominators by making the denominators the same, comparing with one half, or comparing pieces of the same numerator.
Thinking skills	Drawing a fraction model, using a number line, drawing a bar model, making a list in order, comparing with a benchmark, explaining a comparison
Materials	Paper strips (one per child, to fold into halves, quarters and eighths), the three pizza circles from the cut-out page (to cut into quarters), 30 counters, mini whiteboards and markers.
Online set-up	Before the lesson children cut out the 3 pizza circles from the kit page and cut each into 4 equal slices; they fold paper strips on camera and hold up mini whiteboards.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome to the Fraction Frontier! Out here, fractions can be bigger than 1, and two fractions that look different can be the same size."*

"Today we build fractions with bars, line them up on number lines and find out which fraction is really bigger. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about fractions. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Fold a paper strip in half, then in half again, then open it. How many equal parts are there? Colour 3 of them. What fraction of the strip is coloured?	4 equal parts, and 3 of the 4 are coloured, so 3/4 of the strip.
Sneak-peek 2 (thinking): A recipe needs 9 scoops of flour. Each scoop is 1/4 of a cup. How many full cups is that, and how much is left over? Draw it if you like.	2 full cups and 1/4 of a cup left over (9/4 = 2 1/4). Four scoops make one cup: 8 scoops make 2 cups and 1 scoop is left.

DO A good reason sounds like: Four quarters make a whole. Count the quarters in groups of 4: each group is one whole.

4–8 min Warm-up game and quick questions

DO More or less than a half? Thumbs up for more, thumbs down for less, flat hand for exactly a half: 3/8, 5/9, 6/12, 7/10, 2/5, 4/7. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"I will show you a fraction. Thumbs up if it is more than a half, thumbs down if it is less, flat hand if it is exactly a half."*

"Be ready to tell us HOW you know. A hint: what is half of the bottom number?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
1/2 of 18 = ?	9
1/4 of 20 = ?	5
2/3 = ?/12	8
3/4 + 1/8 = ?	7/8
1 − 3/10 = ?	7/10

8–18 min Explore (in pairs, with objects)

SAY *"Each pizza has 4 slices. Ravi's group eats 11 slices. Is that more than 2 pizzas or less? More than 3? Write your guess before you build it."*

DO 1. Cut paper circles into 4 equal slices (quarters).

2. Ravi's group eats 11 slices. Take 11 quarter-slices.

3. Put the slices together to build whole pizzas.

4. Write how many pizzas they ate in two different ways.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK put 4 slices together. That is one whole pizza. How many whole pizzas can you make?

Not sure how to write it: "How many quarters did they eat? How many WHOLE pizzas and how many quarters left over?"

Finished early: what if every pizza had 6 slices and they ate 11 slices?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many pizzas did they eat?	2 whole pizzas and 3 quarters: 11/4 = 2 3/4.
How many slices make exactly 3 pizzas?	12 slices: 12/4 = 3.
What if each pizza had 6 slices and they ate 11?	11/6 = 1 5/6: one whole pizza and 5 sixths.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"7 thirds. 3 thirds make one whole, so I fill one bar: that uses 3. Another 3 fills a second bar. 1 third is left over. So 7/3 is 2 wholes and 1 third: 2 1/3. I can also say 7 ÷ 3 = 2 remainder 1."*

"4 2/3 as thirds: each whole is 3 thirds, so 4 wholes are 12 thirds. Add the 2 thirds: 14 thirds, 14/3."

"3/5 of 30 stickers. The bottom number tells me how many equal units: 5. 5 units are 30, so 1 unit is 6. I need 3 units: 18."

Worked example on the slides	Answer
Improper fraction to mixed number. 7/3: fill whole bars of 3 thirds. 3 + 3 + 1 = 7 thirds. 2 wholes and 1 third.	7/3 = 2 1/3
Fraction of a set. 3/5 of 30 stickers. 5 units = 30, so 1 unit = 6. 3 units = ?	18 stickers

SAY *"The key idea today is: fractions can be more than 1. The denominator tells how many pieces make 1 whole: 7/3 = 2 1/3. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The “your turn” slides are optional: skip them if you are short of time. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Changing a mixed number to an improper fraction by multiplying the whole number by the numerator, or forgetting to add the numerator (3 2/5 written as 6/5 or 15/5).
- Thinking the bigger denominator means the bigger fraction (3/7 > 3/5 "because 7 is bigger than 5").
- Finding a fraction of a set by dividing by the numerator, or stopping at one unit (3/8 of 24 answered 8 or 3).

30–34 min Brain break

DO Fraction stand-up. I call a fraction: stand up if it is more than 1, sit down if it is less than 1, hands on your head if it is exactly 1. 5/4, 3/8, 6/6, 9/5, 7/12, 12/12.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today’s mathematician is Mahavira (India (Karnataka, possibly Mysore), about 800 - about 870 (he worked around 850)). Today you compare and change fractions, and Mahavira's book explained how to work with fractions and split them into unit fractions. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on the story. Try it answer: 1/3 = 2/6, so 1/3 + 1/6 = 2/6 + 1/6 = 3/6 = 1/2. For 3/4: 3/4 = 2/4 + 1/4 = 1/2 + 1/4.

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 4 aloud. Say “Pairs first. Talk before you write.”

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: "How many of these pieces make one whole?"
- Ask: "Draw the bar. How many units? What is one unit worth?"
- Ask: "Can you make the bottom numbers the same? Is it more or less than a half?"

IF STUCK Hint 1: "The denominator tells you how many pieces make one whole."

Hint 2: "For a fraction of a set, divide by the bottom number first, then multiply by the top number."

Hint 3: "To compare, change one fraction so both have the same denominator, or compare each with a half."

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong. The deck has a slide marked ON DEMAND (and in Lesson 5 two): it is not part of the five-minute teaching before page 2. Show it only while children work on page 2, when a child chooses that choice question, and make the offer to the whole group.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ben ate 7/4 pizzas and Ravi ate 1 5/8 pizzas. Who ate more?	Ben. 7/4 = 14/8 = 1 6/8, which is more than 1 5/8.
Workbook Think harder 2: Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?	25 (1 unit = 5, blue = 15, not blue = 5 units = 25).
Workbook Think harder 3: Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12	1/2, 7/12, 2/3, 5/6 (6/12, 7/12, 8/12, 10/12).
Workbook Think harder 4: Who am I? I am an improper fraction. My denominator is 6. I am between 2 and 3. My numerator is an odd number. I am closer to 3 than to 2. Make a list, then cross out the ones that do not fit.	17/6 = 2 5/6. Between 2 and 3: 13/6, 14/6, 15/6, 16/6, 17/6. Odd numerators: 13/6, 15/6, 17/6. 15/6 = 2 1/2 is exactly in the middle and 13/6 is closer to 2, so only 17/6 is closer to 3.
Workbook Go further 1: Mei takes 1/2 of the sweets in a jar. Ali takes 1/3 of the sweets. There are 5 sweets left. (a) How many sweets were in the jar at first? (b) What if 7 sweets were left? (c) Make up your own puzzle with two other fractions taken from a jar and a number of sweets left, and solve it.	(a) 30 sweets: 1/2 + 1/3 = 5/6 are taken, so 1/6 is left; 1/6 = 5, so 6/6 = 30. (b) 42 sweets (1/6 = 7). The jar always holds 6 times the number left. (c) Puzzles will vary: check that the fractions taken add to less than 1 and that the left-over fraction was found first.
Workbook Go further 2: Half of a number is 3 more than a third of the number. What is the number?	18. Half = 3 sixths and a third = 2 sixths, so 1 sixth = 3 and the number is 6 sixths = 18. Check: 9 is 3 more than 6.
Workbook Go further 3: Siti read 2/5 of her book on Monday. On Tuesday she read half of the pages that were left. She still has 45 pages to read. How many pages are in the book?	150 pages. Cut each fifth in half to make tenths: Monday 4 tenths, the rest is 6 tenths, Tuesday 3 tenths, 3 tenths left = 45. 1 tenth = 15, 10 tenths = 150.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Which is more: 2 1/3 pizzas or 9/4 pizzas? Mei says 9/4 because 9 is a big number."*

"Ravi says 3/7 is bigger than 3/5 because 7 is bigger than 5. What would you tell him?"

Discussion question	Listen for
Which is more: 2 1/3 pizzas or 9/4 pizzas? Mei says 9/4 because 9 is a big number. Is she right?	2 1/3 is more. 9/4 = 2 1/4, and both have 2 wholes. A third is bigger than a quarter (sharing one pizza among 3 gives bigger slices than sharing among 4), so 2 1/3 is bigger. Mei looked at the 9 alone, but quarters are small pieces. For Ravi: 3/5 is bigger, because fifths are bigger pieces than sevenths.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Write 23/5 as a mixed number. Then find 3/5 of 25.	4 3/5; 15.

SAY *"You were real fraction explorers today: you built fractions, lined them up and compared them with good reasons."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I drew fraction bars. The hardest part was comparing 3/8 and 3/5. What helped me: drawing two bars of the same length." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone

to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Find a fraction between 2/3 and 3/4. (For example 17/24: 2/3 = 16/24 and 3/4 = 18/24. Another one is 7/10: 2/3 = 40/60, 7/10 = 42/60 and 3/4 = 45/60.)

Answer key (workbook)

Q1 (a) 13 fifths: 13/5 = 2 3/5 (b) 4 3/4 (c) 17/5 (d) 40

Q2 (a) A = 3/4, B = 1 3/4 (7/4), C = 2 2/4 = 2 1/2 (10/4) (b) 9/4 = 2 1/4: one small mark after 2.

Q3 (a) 8 units = 24, 1 unit = 3, 3 units = 9 children. (b) 5 units = 15 children (24 – 9 = 15).

Q4 (a) > (3/4 = 6/8) (b) > (2/3 = 8/12) (c) < (4/9 is less than a half, 5/8 is more) (d) > (fifths are bigger pieces than sevenths) (e) 7/12, 2/3, 5/6

Q5 (a) 3 units = 27, 1 unit = 9, 5 units = 45 sweets. (b) 2 units = 18 sweets are not red.

Q6 1/3 = 4/12 and 3/4 = 9/12. The fractions in between: 5/12, 6/12, 7/12, 8/12. That is 4 fractions (4/12 and 9/12 are equal to 1/3 and 3/4, so they do not count).

Investigation Sharing 24 stickers (What if we change ...? What stays the same?) Easy first step: Draw 24 dots. Circle them in 2 equal groups: how many dots are in one group? That is 1/2 of 24. Now try 3 equal groups. Model answer: (a) 12, 8, 4, 3. For 1/5: 24 does not share equally into 5 (4 each, 4 left over), so it is not a whole number. (b) 1/2, 1/3, 1/4, 1/6, 1/8 and 1/12: exactly the denominators that are factors of 24 (checked by brute force). (c) 60, the smallest common multiple of 2, 3, 4, 5 and 6 (1/2 = 30, 1/3 = 20, 1/4 = 15, 1/5 = 12, 1/6 = 10). Guess and check is fine: 24 fails for 1/5, 30 fails for 1/4, 60 works. Look for the link to factors and common multiples from Lesson 1.

Go further What if each pizza had 8 slices and they ate 19? How many pizzas is that? 2 pizzas and 3 slices: 19/8 = 2 3/8.

Story corner try-it Mahavira: Show that 1/2 = 1/3 + 1/6 by making the denominators the same. Then split 3/4 into 1/2 plus a unit fraction. Answer: 1/3 = 2/6, so 1/3 + 1/6 = 2/6 + 1/6 = 3/6 = 1/2. For 3/4: 3/4 = 2/4 + 1/4 = 1/2 + 1/4.

Sneak-peek puzzles (1) 4 equal parts, and 3 of the 4 are coloured, so 3/4 of the strip. (2) 2 full cups and 1/4 of a cup left over (9/4 = 2 1/4). Four scoops make one cup: 8 scoops make 2 cups and 1 scoop is left.

Lesson 5: Decimal Detectives

Week 3 (first lesson) • Decimals • Workbook: Lesson 5

Objective	Read decimals to 3 places on number lines, compare and order them by place value, round to the nearest whole number or 1 decimal place and work backwards from a rounded number (smallest and largest; rounding to 2 decimal places is only in Go further), and solve money and mass puzzles with bar models.
Thinking skills	Place value to thousandths, zooming in on a number line, ordering, rounding and rounding backwards, systematic listing, bar models, checking every clue
Materials	Digit cards 0 to 9 and a decimal point card for each child (cut from the Digit, point, sign and bracket cards page), a place-value chart (ones, tenths, hundredths, thousandths), mini whiteboards, a metre ruler, play money.
Online set-up	Children cut out the digit cards and a decimal point from the kit page before the lesson and move them on camera. The tutor marks the zoom-in number lines on the slides.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, inquirers! Today you are decimal detectives. Decimals hide in prices, in masses, in race times. A tiny digit after the point can win or lose a race."*

"Our rule today: notice the place, test with a number line, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about decimals. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out or name a child, never announce who is on

which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Ali has 8 ten-cent coins ($0.80). Mei has 7 ten-cent coins and one 5-cent coin ($0.75). Draw the coins. Who has more money?	Ali has more. $0.80 (8 tenths) is more than $0.75 (7 tenths and 5 hundredths). 0.8 is the same as 0.80, and 0.80 is bigger than 0.75.
Sneak-peek 2 (thinking): Write a decimal that is between 3.4 and 3.5. How do you know it fits? Hint: draw a number line from 3.4 to 3.5 with 10 steps.	Any decimal from 3.401 to 3.499, for example 3.45. Write 3.4 as 3.40 and 3.5 as 3.50: 3.45 sits between them.

DO A good reason sounds like: Line up the decimal points and compare place by place. Write extra zeros so every number has the same number of places.

4–8 min Warm-up game and quick questions

DO Target 5. Draw three empty boxes with a decimal point after the first. The tutor turns over digit cards one at a time; write each digit in a box straight away. The number closest to 5 wins. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"Draw a box, a decimal point, then two boxes (three empty boxes, with the point after the first). I will turn over a digit card. Write it in a box straight away. You cannot move it later!"*

"Our target is 5. The number closest to 5 wins. Where would you put a 4? A 9? A 0?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is the 7 worth in 3.475?	7 hundredths (0.07)
Write 6 tenths and 4 thousandths as a decimal.	0.604
Which is bigger: 0.8 or 0.75?	0.8 (0.80 > 0.75)
Round 4.68 to 1 decimal place.	4.7
How much more than $3.05 is $5?	$1.95

8–18 min Explore (in pairs, with objects)

SAY *"You have the cards 2, 5, 8 and a decimal point. How many different numbers do you think you can make? Write your guess before you start."*

DO 1. Take the digit cards 2, 5, 8 and the decimal point card.

2. Make a decimal with all four cards. The point goes between two digits.

3. Make as many different numbers as you can. Write each one down.

4. Put them in order. Which one is closest to 5?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK put the point after the first card. Now swap the digit cards around. Then move the point.

Not sure if you have them all: "How many numbers start with 2? With 5? With 8? Where can the point go?"

Finished early: which number is closest to 50? Which two numbers are closest to each other?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many different numbers did you make?	12: six with 2 decimal places (2.58 … 8.52) and six with 1 decimal place (25.8 … 85.2).
Which is the smallest? The largest?	2.58 and 85.2.
Which number is closest to 5? How do you know?	5.28: it is only 0.28 away. 2.85 is 2.15 away.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Where is the arrow? Between 2.3 and 2.4. I zoom in. That gap is cut into 10 equal steps, so each step is one hundredth, 0.01. The arrow is 6 steps past 2.3, so it is 2.36."*

"Which is bigger, 4.37 or 4.307? I line up the points. Same ones, same tenths. Hundredths: 7 and 0. 7 hundredths is more, so 4.37 is bigger, even though 4.307 has more digits."

"A number with 2 decimal places rounds to 3.8. I look at the number line. The halfway marks are 3.75 and 3.85. 3.75 rounds up to 3.8, so it is the smallest. 3.85 would round up to 3.9, so the largest is 3.84."

Worked example on the slides	Answer
Order: 4.307, 4.37, 4.073, 4.3. Line up the decimal points in a place-value chart. Fill empty places with 0: 4.300, 4.307, 4.370, 4.073. Compare the tenths first, then the hundredths, then the thousandths.	4.073, 4.3, 4.307, 4.37
Rounding backwards. A number with 2 decimal places rounds to 3.8 (to 1 decimal place). The halfway marks are 3.75 and 3.85. What is the smallest it can be? The largest?	Smallest 3.75, largest 3.84 (3.85 rounds to 3.9).

SAY *"The key idea today is: line up the points. Line up the decimal points and compare place by place: ones, tenths, hundredths, thousandths. More digits does not mean bigger. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Comparing decimals by the number of digits (thinking 3.405 > 3.45 because 405 > 45) instead of lining up the decimal points and comparing place by place.
- Reading a number line by counting marks without first working out what one step is worth (0.1, 0.01 or 0.001).
- Rounding backwards: giving 7.45 as the largest number that rounds to 7.4 (it rounds to 7.5), or forgetting 7.35 to 7.39 below 7.4.

30–34 min Brain break

DO Decimal stand-up. The tutor calls a decimal. More than 0.5: stand tall. Less than 0.5: crouch. Exactly 0.5: freeze like a statue.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Simon Stevin (Flanders (born in Bruges, now Belgium), 1548-1620). Today you work with tenths, hundredths and thousandths, which is exactly what Stevin's small book explained. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: 3 + 4/10 + 5/100 = 3.45. Then 3.45 + 2.8 = 6.25.

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 5 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What is one small step worth on this number line? How do you know?"
- Ask: "Line up the decimal points. Which place do you compare first?"
- Ask: "Would this number round up or down? Where is the halfway mark?"

IF STUCK Hint 1: "Write each number in the place-value chart. Fill empty places with 0."

Hint 2: “Count the steps between the two labelled marks. Divide the gap by the number of steps.”

Hint 3: “For bars: take away the extra part first, then share the rest equally.”

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong. The deck has a slide marked ON DEMAND (and in Lesson 5 two): it is not part of the five-minute teaching before page 2. Show it only while children work on page 2, when a child chooses that choice question, and make the offer to the whole group.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Find ALL the digits that the letter A can be: 3.A5 < 3.42 (A is one digit)	0, 1, 2 and 3 (3.45 is more than 3.42)
Workbook Think harder 2: A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?	Smallest 9.50, largest 10.49
Workbook Think harder 3: What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?	2.65 and 2.605
Workbook Think harder 4: Mystery decimal. I have 3 decimal places and I am between 4 and 5. Rounded to 1 decimal place, I am 4.7. My hundredths digit is 2 more than my tenths digit. My digits add up to 25. Who am I?	4.687 (rounds to 4.7, so tenths 6 or 7; tenths 7 would make hundredths 9, 4.79..., too big; so 4.68_ and 4 + 6 + 8 + 7 = 25).
Workbook Go further 1: Make up your own "closest to" puzzle. Choose four digit cards and a decimal point (the point goes between two digits). Pick a target number and find the number with 2 decimal places that is closest to it. Then ask: what if the target changes? Swap puzzles with a partner and check each other’s answers.	Puzzles will vary: check by listing every number that can be made. Model: cards 3, 5, 6, 9 and target 40: the closest is 39.65 (0.35 away); the next is 39.56 (0.44 away). What if the target is 50? Then the closest is 53.69 (3.69 away); the closest below 50 is 39.65, which is 10.35 away.
Workbook Go further 2: Ali, Ben and Siti share $9.20. Ben gets $0.50 more than Ali. Siti gets $1.30 more than Ben. How much does each child get?	Ali $2.30, Ben $2.80, Siti $4.10 (take away $0.50 + $1.80 = $2.30, then $6.90 ÷ 3 = $2.30)

Workbook Go further 3: A jug full of water weighs 1.45 kg. When it is half full, it weighs 0.85 kg. What is the mass of the empty jug?	0.25 kg (half the water is 1.45 − 0.85 = 0.6 kg, all the water is 1.2 kg, jug = 1.45 − 1.2)

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Ben says 3.405 is bigger than 3.45 because 405 is bigger than 45. Who agrees? How can we check?"*

"Why does adding a zero to the end of 3.45 not change it, but putting a zero after the point (3.045) does?"

Discussion question	Listen for
Ben says 3.405 is bigger than 3.45 because 405 is bigger than 45. Is he right?	Ben is wrong. Line up the points: 3.405 and 3.450. The ones and tenths are the same; 0 hundredths is less than 5 hundredths, so 3.405 < 3.45. 3.45 = 3.450 because 0 thousandths adds nothing, but 3.045 moves the 4 into the hundredths place, so it is a different, smaller number.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Write 2.8, 2.08 and 2.088 in order, smallest first. Then round 2.088 to 1 decimal place.	2.08, 2.088, 2.8. 2.088 rounds to 2.1.

SAY *"Today you zoomed in on number lines, lined up the points, and worked rounding backwards. Detectives check every clue, and you did. Well done, inquirers!"*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I wrote the extra zeros. The hardest part was comparing decimals with different numbers of digits. What helped me: adding zeros so they had the same length." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Use the digit cards 1, 3, 5, 7 and a decimal point. Make the number with 2 decimal places that is closest to 40. (37.51) Then make your own closest-to puzzle for a partner.

Answer key (workbook)

Q1 A 6.3, B 6.85, C 4.53, D 4.58, E 0.247. (f) B (6.85 rounds to 7; 6.3 rounds to 6; the others are near 4.5 or 0.2).

Q2 (a) 3.045, 3.054, 3.405, 3.45, 3.5 (b) No. Line up the points: 3.405 and 3.450 have the same tenths, but 0 hundredths is less than 5 hundredths, so 3.405 is smaller.

Q3 (a) Smallest 7.35, largest 7.44 (7.45 rounds up to 7.5). (Moved to the question bank: how many numbers (10) and the digit-sum clue (7.38).)

Q4 (a) 3.69, 3.96, 6.39, 6.93, 9.36, 9.63 (b) 9.63 (c) 6.93 (only 0.07 away) (d) 3.69 and 3.96

Q5 Siti $4.10, Mei $5.30 ($9.40 − $1.20 = $8.20, $8.20 ÷ 2 = $4.10, $4.10 + $1.20 = $5.30; check $4.10 + $5.30 = $9.40).

Q6 (a) 0.4 kg (2.85 − 0.45 = 2.40 kg for 6 cans, 2.40 ÷ 6 = 0.4) (b) 2.05 kg (2.85 − 0.8)

Investigation Between two decimals (Is it always true? Test and explain.) Easy first step: Write 0.35 as 0.350 and 0.36 as 0.360. Now write ONE decimal that sits between them. Model answer: (a) Any of 0.351 to 0.359, for example 0.355. (b) Exactly 9 numbers with 3 decimal places: 0.351, 0.352, ... 0.359 (checked by brute force). (c) Siti is wrong. Between 0.35 and 0.351 there is 0.3505 (4 decimal places), and between 0.35 and 0.3505 there is 0.35025, and so on: there is no "next" decimal after 0.35, you can always find another. Look for place value reasoning (0.35 = 0.350, so 0.351 is a bit more) and for children who write the zeros to compare.

Go further What if the target is 50 instead of 5? Which number with these cards is closest to 50? 52.8 (2.8 away). The next closest, 58.2, is 8.2 away.

Story corner try-it Simon Stevin: Write 3 + 4/10 + 5/100 as a decimal. Then add 2.8 to your answer. Answer: 3 + 4/10 + 5/100 = 3.45. Then 3.45 + 2.8 = 6.25.

Sneak-peek puzzles (1) Ali has more. $0.80 (8 tenths) is more than $0.75 (7 tenths and 5 hundredths). 0.8 is the same as 0.80, and 0.80 is bigger than 0.75. (2) Any decimal from 3.401 to 3.499, for example 3.45. Write 3.4 as 3.40 and 3.5 as 3.50: 3.45 sits between them.

Lesson 6: Angle Turns

Week 3 (second lesson) • Angles • Workbook: Lesson 6

Objective	Estimate and measure angles with a protractor, find every angle in a fan of rays, work out angles between clock hands (30° for each number), turn on the 8-point compass (45° steps), and find an unknown angle in a rectangle corner.
Thinking skills	Estimating with 90° and 180°, measuring, systematic counting, adding and subtracting angles, explaining with turns
Materials	A protractor and a ruler for each child (bring from home: the kit page lists them), the paper clock from the cut-out page with a split pin (or a real clock), a square of paper for checking right angles, mini whiteboards, a compass card on the floor or wall.
Online set-up	Children use their own protractor and ruler and the paper clock cut out from the kit page. Self-view on screen may be mirrored: say turn towards your own right hand, not the screen's left. The 8-point compass is shown on the slides; children stand and turn on camera, with the screen as North.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, inquirers! Today everything turns: you, the hands of a clock, and a robot. An angle tells us how much something has turned."*

"Remember: notice it, test it with a protractor or a paper clock, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about angles and turns. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 2 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Stand up and face the screen. Turn a quarter turn to your right. Then turn another quarter turn the same way. How many quarter turns did you make? Which way are you facing now, compared with the start?	2 quarter turns, which is a half turn. You are facing the opposite way (your back is to the screen).
Sneak-peek 2 (thinking): A square corner is cut into two angles. One angle is 35°. What is the other angle? How do you know?	55°. A square corner is 90°, and 90° – 35° = 55°.

DO A good reason sounds like: A square corner is 90° and a half turn is 180°. Angles side by side add up to the whole turn.

4–8 min Warm-up game and quick questions

DO Robot turns. Stand and face the front (North). The tutor calls a turn; everyone turns and names the direction they now face. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"Everybody stand up and face the front. The front is North. Ready? Quarter turn clockwise! Which way are you facing now?"*

"Back to North. Half turn! Now 45 degrees anticlockwise: just half of a quarter turn. Where are you facing?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
How many degrees in a right angle?	90°
How many degrees in a half turn?	180°
How many right angles make a full turn?	4
Face N. Quarter turn anticlockwise. Which way now?	W
Which is bigger: a 3/4 turn or 250°?	A 3/4 turn (270°)

8–18 min Explore (in pairs, with objects)

SAY *"On a clock, which o'clock times do you think make a right angle between the hands? Write down your guess before you try."*

DO 1. Take a paper clock with two hands.

2. Set it to each o'clock time: 1 o'clock, 2 o'clock … 12 o'clock.

3. Which times make a right angle between the hands? Check with a paper corner.

4. Which time makes a straight line? What is the angle at 1 o'clock?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK put the minute hand on 12. Move only the hour hand, one number at a time.

Not sure if it is a right angle: hold the corner of a square of paper between the hands.

Finished early: what is the angle at 1 o'clock? How can you work it out without a protractor?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which o'clock times make a right angle?	3 o'clock and 9 o'clock.
Which time makes a straight line?	6 o'clock (180°, a half turn).
What is the angle at 1 o'clock? How do you know?	30°: a full turn is 360° and the clock has 12 equal spaces, 360 ÷ 12 = 30.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"I estimate first. This angle is a bit smaller than a right angle, so about 60 or 70 degrees. I put the centre of the protractor on the vertex and the baseline along one arm. That arm points to 0 on the inside scale, so I count up the inside scale: 10, 20 … 65. Not 115: my estimate says it is smaller than 90."*

"A clock face is a full turn, 360 degrees, cut into 12 equal spaces. 360 divided by 12 is 30. At 2 o'clock the hands are 2 spaces apart: 60 degrees. At 10 o'clock I go the shorter way: also 2 spaces, 60 degrees."

"I face North-West. I turn 135 degrees clockwise. Each step on the compass is 45 degrees, so 135 is 3 steps: North, North-East, East. I face East."

Worked example on the slides	Answer
Measure with a protractor. Estimate: the angle is smaller than a right angle. Centre on the vertex, baseline along one arm. Count up from the 0 on that arm: which number?	65° (not 115°: the estimate says less than 90°).

Compass turns: 45° steps. Face NW. Turn 135° clockwise. 135° = 3 steps of 45°. Count 3 steps clockwise: N, NE, …	E.

SAY *"The key idea today is: count the turn. A full turn is 360°: 4 right angles, 8 compass steps of 45°, 12 clock spaces of 30°. Estimate first, then measure. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Reading the wrong protractor scale (writing 140° for a 40° angle) instead of starting from the 0 on the arm and checking with an estimate.
- Counting the numbers on the clock instead of the spaces between them, or measuring the bigger way round (210° instead of 150° at 7 o'clock).
- Compass turns: mixing up clockwise and anticlockwise, or counting the starting direction as the first 45° step.

30–34 min Brain break

DO Compass dance. Stand facing North. Jump a quarter turn clockwise on each clap: N, E, S, W, N. Then half turns, then 45° steps saying each direction.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Eratosthenes (Cyrene (now Libya) and Alexandria, Egypt; ancient Greek world, about 276 BC - about 194 BC). Today you measure angles and turns, and Eratosthenes used the angle of a shadow to measure the whole Earth. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: 360 divided by 10 = 36, so 36 angles. (Eratosthenes had about 7.2 degrees, and 360 divided by 7.2 = 50.)

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 6 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “Is this angle bigger or smaller than a right angle? So which number on the protractor makes sense?”
- Ask: “How many spaces apart are the hands? How many degrees is one space?”
- Ask: “How many 45° steps is that turn? Which way: clockwise or anticlockwise?”

IF STUCK Hint 1: “Estimate first: compare with the corner of a page (90°).”

Hint 2: “Draw the clock or the compass and count the spaces or steps with your pencil.”

Hint 3: “For a corner of a square or rectangle: the parts add up to 90°. Take away the parts you know.”

DO Choosing: after question 2 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Siti faces SE. She turns 225° clockwise. Which direction does she face now?	N (225° is 5 steps of 45°: S, SW, W, NW, N)
Workbook Think harder 2: What is the angle between the clock hands at 10 o’clock? At 2 o’clock? What do you notice?	60° and 60°. Both are 2 spaces apart the shorter way.
Workbook Think harder 3: How many degrees does the minute hand turn from 7:15 to 7:50?	210° (7 jumps of 5 minutes, and each jump is one clock space of 30°)
Workbook Think harder 4: A robot starts facing N. Ben wants it to turn from N to W using two DIFFERENT cards from: 45° clockwise, 90° clockwise, 135° anticlockwise, 180°. Find ALL the pairs that work.	Two ways: 45° clockwise and 135° anticlockwise; or 90° clockwise and 180°.
Workbook Go further 1: Make up your own clock-angle puzzle. Choose a time that is NOT an o’clock time (half past is easiest), work out the angle between the hands, and write your puzzle as "What is the angle at ...?". Draw the clock to check your answer. Then find a second time with the same angle.	Puzzles will vary. Model: 9:30 is 105° (the hour hand is halfway between 9 and 10, 3 and a half spaces, 3 × 30° + 15°). Another time with the same angle is 2:30.

Workbook Go further 2: The corner of a square is cut into 3 angles. The middle angle is twice the smallest. The biggest is 3 times the smallest. Find all three angles.	15°, 30° and 45° (6 equal parts make 90°, so one part is 15°)
Workbook Go further 3: Two squares share one corner O and overlap, like the picture. The angle covered by both squares together (from the outside side of one square to the outside side of the other) is 160°. What is the angle where the squares overlap?	20° (90° + 90° = 180°, and 180° − 160° = 20°)

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Is turning 270° clockwise the same as turning 90° anticlockwise? Test it: stand up and do both."*

"Ben says the angle at 3 o'clock is smaller than at 9 o'clock because 3 is smaller than 9. Who agrees?"

Discussion question	Listen for
Is turning 270° clockwise the same as turning 90° anticlockwise? How do you know?	Yes: 270° clockwise and 90° anticlockwise make a full turn (360°) together, so you end up facing the same way. Ben is wrong: at 3 o'clock and at 9 o'clock the hands are 3 spaces apart the shorter way, so both angles are 90°.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Face N. Turn 135° clockwise. Which way do you face? What is the angle between the clock hands at 5 o'clock?	SE. 150° (5 spaces of 30°).

SAY *"Today you estimated before you measured, counted spaces on a clock and steps on a compass, and found every angle in a fan. Inquirers check with an estimate. Well done!"*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start

of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I estimated each angle before I measured. The hardest part was reading the protractor scale. What helped me: starting from the 0 that lines up with one arm." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Find all the o'clock times when the angle between the hands is more than 90° (the smaller angle). (4, 5, 6, 7 and 8 o'clock.) Then: at 6:30 the hands are not on top of each other. Why not?

Answer key (workbook)

Q1 a 40°, b 115°, c 75°, d 150° (allow 2° either way; estimates within about 20° are good). b and d are bigger than a right angle.

Q2 (a) 6 angles (b) ∠AOB 30°, ∠BOC 60°, ∠COD 45°, ∠AOC 90°, ∠BOD 105°, ∠AOD 135° (c) ∠AOC

Q3 (a) 120° (b) 150°

Q4 (a) N (b) E

Q5 ∠a = 32° (90° – 58°)

Q6 (a) SW (N → E → NE → SW)

Investigation Splitting a corner (Find all the ways.) Easy first step: A square corner is 90°. Draw it and split it into three EQUAL angles. How big is each one? Model answer: (a) Seven sets: 10, 10, 70; 10, 20, 60; 10, 30, 50; 10, 40, 40; 20, 20, 50; 20, 30, 40; 30, 30, 30 (checked by brute force). (b) A good method: fix the smallest angle, then list the other two in order. Counting in units of 10° helps: three whole numbers that add up to 9. (c) Three sets: 30, 30, 120; 30, 60, 90; 60, 60, 60. What stays the same: the three angles always add up to the whole turn (90° or 180°). Low floor: any three angles that add to 90°. Look for children who realise a list needs a system.

Go further What if the clock shows half past three (3:30)? Is the angle still 90°? Why not? No. It is 75°: at 3:30 the hour hand has moved halfway from 3 to 4, which is 15° past the 3. The minute hand points at 6 (180°) and the hour hand at 105°.

Story corner try-it Eratosthenes: A whole turn is 360 degrees. If a shadow made an angle of 10 degrees, how many angles that size fit all the way round? Answer: 360 divided by 10 = 36, so 36 angles. (Eratosthenes had about 7.2 degrees, and 360 divided by 7.2 = 50.)

Sneak-peek puzzles (1) 2 quarter turns, which is a half turn. You are facing the opposite way (your back is to the screen). (2) 55°. A square corner is 90°, and 90° – 35° = 55°.

Lesson 7: Area Architects II

Week 4 (first lesson) • Area and Perimeter • Workbook: Lesson 7

Objective	Find the area of a composite figure by splitting it into rectangles or by taking away a missing piece, find its perimeter by pushing the steps out to a rectangle, find every rectangle with a given area and compare their perimeters, and explain what changes when a shape is cut and rearranged.
Thinking skills	Splitting and taking away, making a list in order, looking for a pattern, explaining why (area stays the same, perimeter can change)
Materials	1 cm square tiles or 24 paper squares per child (cut the Square grid page into squares), 1 cm grid paper, scissors, a 30 cm ruler, mini whiteboards, a paper strip 8 squares by 2 squares for cutting.
Online set-up	Children cut 24 paper squares and some 1 cm grid strips from the Square grid page before the lesson and build on camera; the tutor draws on the grid figures on the slides.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome, Area Architects! Architects must know two things about every room: how much floor it covers, and how far it is all the way round."*

"Today we find areas and perimeters of tricky shapes WITHOUT counting every square, and we test a big idea: does the same area always give the same perimeter? Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about area and perimeter. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Look at the picture: a 5 by 3 rectangle of squares with one corner square cut out. Count the squares that are left. How many squares is the area?	14 squares (the full 5 by 3 rectangle is 15 squares, and 1 is cut out).
Sneak-peek 2 (thinking): The perimeter is the distance all the way round the edge. Is the perimeter of this shape longer than, shorter than or the same as the perimeter of the full 5 by 3 rectangle? Say why.	The same: 16 units for both. Push the cut-out corner back out. The two edges of the notch are just as long as the two edges of the corner they replace.

DO A good reason sounds like: For area, count the squares or take away the piece that is missing. For perimeter, push the corner back out to a rectangle.

4–8 min Warm-up game and quick questions

DO Area or Perimeter? The tutor reads a job. Hold up A if it covers the inside, P if it goes round the edge. Then explain why. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"I will read out a job. If the job needs the space INSIDE, hold up A for area. If it goes ROUND THE EDGE, hold up P for perimeter."*

"Be ready to explain: "It covers the floor, so it is area." "It goes round the outside, so it is perimeter."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Area of a rectangle 7 cm by 6 cm?	42 cm²
Perimeter of a square with sides of 9 cm?	36 cm
A rectangle has area 48 cm² and length 8 cm. Breadth?	6 cm
Perimeter of a rectangle 12 cm by 5 cm?	34 cm
A square has perimeter 28 cm. Its area?	49 cm²

8–18 min Explore (in pairs, with objects)

SAY *"You have 12 square tiles. How many different rectangles do you think you can make with all 12? Will they all have the same perimeter? Write your guess in the corner of your whiteboard."*

DO 1. Take 12 square tiles. Each is 1 cm by 1 cm.

2. Use all 12 to make a rectangle. Write its length, breadth and perimeter.

3. Make every different rectangle you can.

4. Which has the longest perimeter? The shortest? What do you notice?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK start with one long row of 12. Then try 2 rows. Then 3 rows.

Not sure if a rectangle is new: "Is it the same as one you have, just turned round?"

Finished early: try 16 tiles. Which rectangle has the smallest perimeter now?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many different rectangles did you make?	3: 1 by 12, 2 by 6 and 3 by 4 (turning one round does not make a new one).
Which has the longest perimeter? The shortest?	1 by 12 has 26 cm; 3 by 4 has 14 cm.
What do you notice?	All have area 12 cm², but the perimeters differ. Long and thin is biggest; closest to a square is smallest.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"An L-shape. I need every side. The bottom is 8 cm and the top part is 5 cm, so the step across must be 8 − 5 = 3 cm. The left is 6 cm and the right is 4 cm, so the step down is 6 − 4 = 2 cm."*

"For the area I can split it into a 5 by 6 rectangle and a 3 by 4 rectangle: 30 + 12 = 42. Or I can draw the big 8 by 6 rectangle and take away the missing 3 by 2 piece: 48 − 6 = 42. Same answer, so I trust it."

"For the perimeter, I push the two step sides out. The 3 cm piece slides up to the top and the 2 cm piece slides across to the right. Now it is an 8 by 6 rectangle, so the perimeter is 8 + 6 + 8 + 6 = 28 cm."

"Area 20: I list the pairs in order. 1 × 20, 2 × 10, 3 does not work, 4 × 5, and 5 × 4 is the same as 4 × 5, so I stop. Perimeters 42, 24, 18. Long and thin is biggest; close to a square is smallest."

Worked example on the slides	Answer

Area of an L-shape: split or take away. An 8 cm by 6 cm rectangle with a 3 cm by 2 cm piece missing from one corner. Split: 5 × 6 + 3 × 4. Take away: 8 × 6 – 3 × 2.	42 cm² both ways.
Perimeter: push the steps out. The same L-shape. Two sides have no labels. Slide the step sides out to the edges. It becomes an 8 cm by 6 cm rectangle.	28 cm, the same as the rectangle.

SAY *"The key idea today is: split, take away, push out. Area: split into rectangles or take the missing piece away. Perimeter: push the steps out to make a rectangle. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding the given sides only and forgetting the unlabelled sides when finding the perimeter of an L-shape (or labelling a missing side with the wrong length because they did not use the opposite side).
- Thinking that the same area means the same perimeter, or that cutting a corner off a rectangle makes the perimeter smaller.
- Finding the area of a border by working out 2 × (length + breadth) and forgetting the 4 corner squares, instead of big rectangle take away the inside.

30–34 min Brain break

DO Body shapes! "Long and thin": stretch your arms out wide. "Square": hug yourself tight. Which shape has the shorter way round?

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Euclid (Alexandria, Egypt (Greek-speaking world), about 325 BC - about 265 BC (dates are estimates)). Today you cut and rearrange shapes and explain what happens to area, which is the idea in Euclid's rule for parallelograms. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: Rectangle: 8 x 5 = 40 square cm. The slanted parallelogram has the same base and height, so its area is also 40 square cm. Its slanted sides are longer than 5 cm, so it has the longer way round (a bigger perimeter).

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 7 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check

page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which side is opposite this one? How can that help you find the missing length?"
- Ask: "Can you find the area another way to check?"
- Ask: "You cut the shape and moved a piece. What stayed the same? What changed?"

IF STUCK Hint 1: "Mark every side with its length before you start. Use the opposite sides."

Hint 2: "Draw the big rectangle round the shape. What is missing?"

Hint 3: "Push the steps out. Does the outline become a rectangle?"

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask "How did you decide?" and "Is there another way?" rather than marking it right or wrong. The deck has a slide marked ON DEMAND (and in Lesson 5 two): it is not part of the five-minute teaching before page 2. Show it only while children work on page 2, when a child chooses that choice question, and make the offer to the whole group.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?	1 by 30 (62 cm), 2 by 15 (34 cm), 3 by 10 (26 cm), 5 by 6 (22 cm). Smallest: 5 cm by 6 cm, 22 cm.
Workbook Think harder 2: A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?	1 by 9 (9 cm²), 2 by 8 (16 cm²), 3 by 7 (21 cm²), 4 by 6 (24 cm²), 5 by 5 (25 cm²). Largest: the 5 cm by 5 cm square, 25 cm².
Workbook Think harder 3: A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.	Area 96 − 4 × 9 = 60 cm². Perimeter 40 cm, the same as the rectangle: each cut takes away 3 + 3 and adds 3 + 3.
Workbook Think harder 4: Three squares with sides 2 cm, 3 cm and 4 cm stand side by side in a row on the same line, smallest on the left. Find the area and the perimeter of the figure.	Area 4 + 9 + 16 = 29 cm². Perimeter 26 cm: push the steps out to a 9 cm by 4 cm rectangle.

Workbook Go further 1: Make up your own fencing puzzle like Ravi's: choose a length of fence (a whole number of metres, more than 20) and say whether one side is a wall. Find the largest area the pen can have, and prove it by listing every whole-metre pen. Swap puzzles with a partner and check each other's lists.	Puzzles will vary: check that every whole-metre pen is listed. Model: 30 m of fence, no wall: the sides add to 15, and 7 m by 8 m gives the largest area, 56 m². With a wall on one side: 7 m, 16 m, 7 m (or 8 m, 14 m, 8 m) both give the largest area, 112 m².
Workbook Go further 2: A garden is 10 m long and 8 m wide. Two paths 1 m wide cross it: one goes along its length and one goes across its width. What area of the garden is NOT path?	63 m². Slide the paths to the edges: the grass makes a 9 m by 7 m rectangle. (Or 80 − 10 − 8 + 1 = 63, because the square where the paths cross was counted twice.)
Workbook Go further 3: A rectangle 9 cm long and 4 cm wide is cut into 2 pieces. The pieces fit together exactly to make a square. How long is a side of the square? Did the perimeter go up or down?	Area 36 cm², so the side is 6 cm. The perimeter went down from 26 cm to 24 cm.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Ben says a bigger area always means a bigger perimeter. Can you find two rectangles that prove him wrong?"*

"When we cut a shape and move the pieces, which stays the same, area or perimeter? Why?"

Discussion question	Listen for
Ben says: "A bigger area always means a bigger perimeter." Do you agree? Use rectangles to test it.	Ben is wrong. A 1 cm by 12 cm rectangle has area 12 cm² and perimeter 26 cm, but a 4 cm by 4 cm square has a bigger area (16 cm²) and a smaller perimeter (16 cm). When we cut and move pieces, nothing is added or lost, so the area stays the same; the outline changes, so the perimeter can change.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer

A 7 cm by 5 cm rectangle has a 2 cm by 2 cm square cut from one corner. Find the area and the perimeter of the shape that is left.	Area 35 − 4 = 31 cm². Perimeter 24 cm, the same as the rectangle (push the corner back out).

SAY *"Today you were real architects: you split shapes, took pieces away and pushed steps out instead of counting every square."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I wrote the lengths on the picture. The hardest part was finding the missing lengths on the staircase. What helped me: writing every length I could work out on the picture." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Ravi has 24 m of fencing for a rectangular pen with whole-metre sides. One side is a wall and needs no fence. What is the largest area he can fence? (72 m²: 6 m, 12 m, 6 m.)

Answer key (workbook)

Q1 (a) The step across is 5 cm (9 − 4) and the step down is 3 cm (7 − 4). (b) Split: 4 × 7 + 5 × 4 = 28 + 20 = 48 cm². Take away: 9 × 7 − 5 × 3 = 63 − 15 = 48 cm². (c) 4 + 3 + 5 + 4 + 9 + 7 = 32 cm, the same as a 9 cm by 7 cm rectangle.

Q2 Table: 1 by 24, perimeter 50 cm; 2 by 12, 28 cm; 3 by 8, 22 cm; 4 by 6, 20 cm (4 rectangles; 6 by 4 is the same as 4 by 6). (a) Largest: 1 cm by 24 cm (50 cm). Smallest: 4 cm by 6 cm (20 cm). (The 36 cm² question, answer: 6 cm by 6 cm, a square with perimeter 24 cm, is in the question bank.)

Q3 (a) 6 cm² (b) Smallest: a 2 by 3 rectangle (the only shape with the smallest perimeter): 10 cm. Largest: for example a row of 6 squares, or any shape with no 2 by 2 block of squares in it: 14 cm.

Q4 (a) 34 cm. Push the steps out: they make an 11 cm by 6 cm rectangle, and 11 + 6 + 11 + 6 = 34. (b) 3 × 2 + 7 × 2 + 11 × 2 = 6 + 14 + 22 = 42 cm².

Q5 (a) A 4 cm by 4 cm square (the two 4 by 2 pieces stacked). (b) Rectangle: area 16 cm², perimeter 20 cm. Square: area 16 cm², perimeter 16 cm. (c) No. The area stayed the same, but the perimeter changed from 20 cm to 16 cm.

Q6 (a) 10 cm by 8 cm (b) $10 \times 8 - 8 \times 6 = 80 - 48 = 32$ cm² (c) Her 4 strips ($8 + 8 + 6 + 6 = 28$ cm²) leave out the 4 corner squares of the frame: $28 + 4 = 32$ cm².

Investigation Bigger area, bigger perimeter? (Is it always true? Test and explain.) Easy first step: Draw a 4 cm by 4 cm square on the Square grid. Write its area and its perimeter. Model answer: (a) 1 by 11: area 11 cm², perimeter 24 cm. 4 by 4: area 16 cm², perimeter 16 cm. The square has the bigger area but the smaller perimeter, so Ali is wrong. (b) Two sides add up to 12 cm: 1 by 11 (11 cm²), 2 by 10 (20), 3 by 9 (27), 4 by 8 (32), 5 by 7 (35), 6 by 6 (36). Six rectangles (checked by brute force). (c) The 6 by 6 square has the biggest area. For the same perimeter, the closer to a square, the bigger the area. Look for an ordered list, and for children who see that the pair of sides adds up to half the perimeter.

Go further What if you use 24 tiles? Make every rectangle. Which has the smallest perimeter? 4 rectangles: 1×24 (perimeter 50 cm), 2×12 (28 cm), 3×8 (22 cm) and 4×6 (20 cm). The 4 by 6 has the smallest perimeter: it is the closest to a square.

Story corner try-it Euclid: A rectangle is 8 cm long and 5 cm high. Slide its top side along to make a slanted shape with the same base and height. What is the area of each? Which has the longer way round? Answer: Rectangle: 8 x 5 = 40 square cm. The slanted parallelogram has the same base and height, so its area is also 40 square cm. Its slanted sides are longer than 5 cm, so it has the longer way round (a bigger perimeter).

Sneak-peek puzzles (1) 14 squares (the full 5 by 3 rectangle is 15 squares, and 1 is cut out). (2) The same: 16 units for both. Push the cut-out corner back out. The two edges of the notch are just as long as the two edges of the corner they replace.

Lesson 8: Symmetry and Nets

Week 4 (second lesson) • Symmetry and Nets • Workbook: Lesson 8

Objective	Decide whether a shape made of 6 squares folds into a cube by folding it in your head, find which faces of a net are opposite (and use it for dice), complete a figure so that a given line is a line of symmetry, find all the lines of symmetry of a shape, and shade squares to make a pattern symmetric.
Thinking skills	Visualising (folding in your head), making a list in order, testing every case, explaining why
Materials	Squared paper (2 cm squares), scissors, sticky tape, 6 paper squares per child (cut from the Square grid page), a dice, a small mirror (optional), mini whiteboards, coloured pencils.
Online set-up	Children cut out 6 paper squares from the Square grid page, and the cube net, before the lesson and fold on camera; the tutor clicks to reveal each fold on the slides and shades the grids.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome, Shape Detectives! Today we fold things: paper into cubes, and pictures in half in our heads."*

"Some shapes made of 6 squares fold into a perfect cube and some do not. By the end you will be able to tell which, without cutting. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about symmetry and nets. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Write a big capital letter H. Draw every fold line where the two halves would match exactly. How many fold lines can you find?	2 fold lines: one up-down line through the middle and one across the middle. (Plain block H.)
Sneak-peek 2 (thinking): Could a straight row of 6 squares fold into a cube? Cut 6 paper squares and try, if you like. Explain.	No. The first 4 squares wrap round as walls. The 5th lands on top of the 1st and the 6th on top of the 2nd, so nothing makes a lid or a base.

DO A good reason sounds like: A net needs one base, one lid and four walls with nothing overlapping. A line of symmetry is a fold where both halves match exactly.

4–8 min Warm-up game and quick questions

DO Fold It in Your Head. A letter made of squares appears. Show with your fingers how many lines of symmetry it has, then show where the fold goes. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"I will show a letter made of squares. Imagine folding it so the two halves match exactly. How many different ways can you fold it? Show me with your fingers."*

"Then tell me where the fold line goes: up and down, or across?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
How many faces does a cube have?	6
How many edges?	12
How many corners?	8
On a dice, 1 is opposite 6 and 2 is opposite 5. What is opposite 3?	4
How many ways can you fold a rectangle (not a square) so the halves match?	2

8–18 min Explore (in pairs, with objects)

SAY *"You have 6 squares. Do you think EVERY shape you make with them will fold into a cube? Write yes or no in the corner of your whiteboard."*

DO 1. Take 6 paper squares.

2. Join them edge to edge (whole sides touching) and tape them.

3. Try to fold your shape into a cube: no gaps, no overlaps.

4. Draw every shape that works. Did any fail? Why?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK start with 4 squares in a row, which wrap round to make the walls. Where can the other two go?

Not sure if it is new: "Is it the same as one we have, just turned or flipped over?"

Finished early: find a shape that does NOT fold. Why does it fail?

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many different nets did your group find?	There are 11 altogether (turned or flipped copies are the same).
What made a shape fail?	Two squares landed on the same face: for example a 2 by 2 block, 5 in a row, or two squares closing the same end.
Do all nets have 4 squares in a row?	No. Some have rows of 3 (2 rows of 3), and the staircase has no more than 2 in a row.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"I choose a base: a square with lots of neighbours. The squares touching it fold up and become walls: left, right and back. This square is joined to the right wall, so when the right wall stands up it swings round to the front. The last square folds over the top: the lid. One base, four walls, one lid. It works."*

"This one: base in the middle. Two walls go up, front and back. This square hangs from the back wall and swings round to the left. But this other one hangs from the front wall and ALSO swings to the left. Two squares in the same place, and nothing on the right. Not a net."

"Opposite faces: in a straight line of 3, the two ends are opposite, because the middle one folds between them. At the two ends of a step, the ends are opposite too. Then the last two letters must be the last pair."

Worked example on the slides	Answer

Fold it in your head. Shade a base. The squares touching it fold up as walls. A square joined to a wall swings round with it. Is there one base, one lid and four walls?	Yes: it is a net of a cube. If two squares land in the same place, it is not.
Opposite faces. In a straight line of 3 squares, the ends are opposite. The two ends of a step are opposite. The last two faces are the last pair.	On a dice, opposite faces add up to 7.

SAY *"The key idea today is: fold it in your head. A net works when it has one base, one lid and four walls with nothing overlapping. In a line of 3, the ends are opposite. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Thinking any 6 squares make a cube, and ticking shapes with a 2 by 2 block, 5 squares in a row, or two squares that close the same end.
- Saying the two squares next to each other in a net are opposite, instead of the squares with one square between them.
- Saying a rectangle has 4 lines of symmetry (including the diagonals), or completing a symmetric figure by copying (sliding) the half instead of flipping it.

30–34 min Brain break

DO Mirror partner! The tutor moves slowly; you are the mirror. Raise your right hand when the tutor raises the left.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Emmy Noether (Germany (born in Erlangen), 1882-1935). Today you explore symmetry, and Emmy showed that symmetry is useful as well as beautiful. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: A square has 4 lines of symmetry (2 through the middles of opposite sides and 2 along the diagonals). A rectangle that is not a square has 2 (through the middles of opposite sides).

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 8 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check

page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which square is your base? Which ones become walls?"
- Ask: "Which face lands on top of this one? Is anything already there?"
- Ask: "How many squares is this corner from the line? Where is its partner?"

IF STUCK Hint 1: "Shade the base. Fold the walls up one at a time with your finger."

Hint 2: "Look for a straight line of 3 squares. The ends are opposite."

Hint 3: "Count the squares from the dashed line to each corner. Put a dot the same number of squares on the other side."

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask "How did you decide?" and "Is there another way?" rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?	22 squares: 2 faces of 3 by 2 (6 each), 2 faces of 3 by 1 (3 each) and 2 faces of 2 by 1 (2 each). 12 + 6 + 4 = 22.
Workbook Think harder 2: Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?	Bottom 4 (3 + 4 = 7). Back 2 (5 + 2 = 7).
Workbook Think harder 3: A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?	2. All six faces add up to 1 + 2 + 3 + 4 + 5 + 6 = 21, and 21 − 19 = 2.
Workbook Think harder 4: There are 5 different shapes made of 4 squares joined edge to edge (a line, a square, an L, a T and a zigzag). Which one has exactly 1 line of symmetry?	Only the T. The line has 2, the square has 4, the L and the zigzag have 0.

Workbook Go further 1: Design your own dice. Draw a net of a cube that is not one of the nets in this lesson, and write the numbers 1 to 6 on it so that opposite faces add up to 7. Name each pair of opposite faces to prove it works. Then rub out two numbers and challenge a partner to put them back.	Designs will vary: check each pair of opposite faces on the net adds up to 7. Model: a row of four squares 2, 1, 5, 6 with a 3 above the 1 and a 4 below the 1. Opposite pairs: 2 and 5, 1 and 6 (first and third, second and fourth in the row), and 3 and 4 (above and below).
Workbook Go further 2: In how many ways can you shade exactly 2 squares of a 3 by 3 grid so that the up-down line through the middle is a line of symmetry?	6. Both squares in the middle column: 3 ways. One square in each side column, in the same row: 3 ways.
Workbook Go further 3: Shade exactly 5 squares of a 3 by 3 grid so that the pattern has 4 lines of symmetry. There are 2 ways. Find both.	The middle square and the 4 corners (an X), or the middle square and the 4 edge-middle squares (a plus).

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Mei says every shape made of 6 squares folds into a cube. Ben says a shape with 4 squares in a row always does. Who is right?"*

"How can you tell two faces are opposite without folding?"

Discussion question	Listen for
Mei says: "Every shape made of 6 squares folds into a cube." Ben says: "If a shape has 4 squares in a row, it always folds into a cube." Who is right?	Neither is right. A 2 by 3 rectangle has 6 squares but does not fold into a cube (Mei is wrong). A row of 4 with both extra squares on the same side does not fold either, because both close the same end (Ben is wrong). In a straight line of 3 squares, the two ends are opposite.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
This dice net shows 1, 3 and 2. Which number goes in the square marked "?"	4. The "?" square is opposite the 3 (the ends of a step), and 3 + 4 = 7.

SAY *"You folded cubes in your head today, and you found the hidden partners in symmetric shapes."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I made the net with paper. The hardest part was folding the net in my head. What helped me: making it with paper and tape." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

How many different nets of a cube are there? (11, if turned or flipped copies count as the same.) Draw as many as you can on squared paper.

Answer key (workbook)

Q1 (a) 3 shapes: A, C and E; one shape that does not fold, with a reason, for example: B has two squares on the same side of the row of 4, so both close the same end. D has 5 squares in a row: the 5th lands on the 1st. F has a 2 by 2 block: 4 squares meet at one corner, but only 3 faces of a cube meet at a corner.

Q2 Opposite A: E. Opposite B: D. Opposite C: F.

Q3 (a) The long row reads 3, 5, 4, 2 (left to right) and the square below the right end is 6. (b) Top row 5, 4; middle row 1, 2; bottom row 3, 6. (c) 1 and 6 are the ends of a step, so they are opposite and add up to 7.

Q4 (a) Rocket: flip each corner across the dashed line, the same number of squares on the other side: shoulders 2 squares to the right of the line, fin tips 4 squares to the right at the bottom, bottom edge ends 1 square right of the line, notch on the line, then join. (b) Fish: flip each corner below the line, the same number of squares: tail tip 2 below, then 1 below, back 3 below, 2 below, nose on the line.

Q5 (a) 1 (up-down) (b) 2 (up-down and across) (c) 4 (up-down, across and both corner-to-corner) (d) 0

Q6 2 ways: shade A2 (the up-down line becomes a line of symmetry) or shade D3 (the across line becomes a line of symmetry). No other square works.

Investigation A row of four (Is it always true? Test and explain.) Easy first step: Cut 6 squares from the Square grid page. Make the net in the picture (a row of 4 squares with one square above it and one below). Fold it. Does it make a cube? Model answer: Mei is right. All 16 ways to place the top square and

the bottom square give a cube (checked by folding every case on a computer). The row of 4 wraps round as four walls; the square above becomes the lid and the square below becomes the base, wherever they are along the row. (c) With both extra squares above the row, none of the 6 ways folds into a cube: both squares fold over to become a lid, so they land on top of each other and the base is empty. Look for children who test the end positions too, and who use wall, lid and base in their reason.

Go further What if the longest row has only 3 squares? Can the shape still fold into a cube? Find one. Yes. 4 of the 11 cube nets have a longest row of 3 squares. One example: two rows of 3 squares, where the second row starts under the last square of the first row. (One net, the staircase, has no row longer than 2.)

Story corner try-it Emmy Noether: How many lines of symmetry does a square have? How many does a rectangle that is not a square have? Answer: A square has 4 lines of symmetry (2 through the middles of opposite sides and 2 along the diagonals). A rectangle that is not a square has 2 (through the middles of opposite sides).

Sneak-peek puzzles (1) 2 fold lines: one up-down line through the middle and one across the middle. (Plain block H.) (2) No. The first 4 squares wrap round as walls. The 5th lands on top of the 1st and the 6th on top of the 2nd, so nothing makes a lid or a base.

Lesson 9: Data Detectives

Week 5 (first lesson) • Tables and line graphs • Workbook: Lesson 9

Objective	Read values and trends from line graphs (using the scale), find the biggest increase or decrease, find missing values in tables and graphs from totals and patterns, compare two lines on one graph, and explain why a graph with a different scale can mislead.
Thinking skills	Reading scales, comparing changes, working from totals, reasoning about how a graph is drawn
Materials	Mini whiteboards, a ruler and two coloured pencils per child, the blank graph grids from the kit page (or squared paper), the queue table for the explore task.
Online set-up	Children use the printed Graph grids page or squared paper and hold it up to the camera; the tutor plots on the slide graphs while children call out where each point goes.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, Data Detectives! A graph is a picture of numbers. A good detective reads it carefully, and also checks whether the picture is telling the truth."*

"Today we read line graphs and tables, find numbers that are missing, and catch a graph that tries to trick us. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about line graphs. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 2 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out or name a child, never announce who is on

which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): A stall sold cups of juice: Monday 30, Tuesday 45, Wednesday 25. Which day sold the most? Which day sold the fewest? Draw a small bar for each day if you like.	Tuesday sold the most (45). Wednesday sold the fewest (25).
Sneak-peek 2 (thinking): A line graph has gridlines at 0, 50, 100, 150 and 200. A point is exactly halfway between the 100 line and the 150 line. What number does it show?	125. Halfway between 100 and 150 is 100 + 25.

DO A good reason sounds like: Read the scale first: what is each line worth? Halfway between two lines is half of the step.

4–8 min Warm-up game and quick questions

DO Which graph tells the story? Match four stories (a bath filling, a hawker centre from 10 am to 3 pm, a candle burning, a piggy bank nobody touches) to four line shapes, and say why. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"Here are four tiny graphs with no numbers at all. I will read a story. Point to the graph that tells the story, and be ready to say why."*

"Listen: a bath filling up with water. Which line? Why not the one that goes up and then down?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
75 – 48 = ?	27
On a scale in 10s, what is halfway between 40 and 50?	45
38 + 47 + 15 = ?	100
Four numbers total 200. Three are 45, 60 and 50. The fourth?	45
Bigger change: 120 to 95, or 60 to 90?	60 to 90 (30)

8–18 min Explore (in pairs, with objects)

SAY *"You will plot the queue at a chicken rice stall from 11:00 to 2:00. Before you plot: at what time do you think the queue is longest? Write your guess at the top of your grid."*

DO 1. Look at the table: people in the queue at a chicken rice stall every half hour.

2. Choose a scale for the side of your grid (how much is each line worth?).

3. Plot each point and join the points with straight lines.

4. Write three things your graph tells you.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK the biggest number is 22. If each line is 2 people, how many lines do you need to reach 22?

Points in the wrong place: put your finger on the time, go straight up, then check the number on the side.

Finished early: between which two times did the queue grow the most? Show it on your graph without working out every change.

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
When was the queue longest?	12:30, with 22 people.
Between which two times did the queue grow the most?	11:30 to 12:00: up 9 people (the steepest line going up).
When is the best time to buy chicken rice? Why?	2:00 (3 people), or 11:00 (4 people): the shortest queues.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"First the title and the axes: books borrowed, Monday to Friday. Then the scale: each line on the side is worth 10. Tuesday's point is on the 40 line, so 40. Monday's point is halfway between 20 and 30, so 25."*

"Which day had the biggest change? I list each change: Monday to Tuesday up 15, Tuesday to Wednesday down 5, Wednesday to Thursday up 25, Thursday to Friday down 10. The biggest increase is 25, and on the graph it is the steepest line going up."

"A table with missing numbers: Ali's row must add up to 61. 61 − 26 = 35. I fill the rows first, then the columns. Last, I check: the column totals and the row totals must give the same grand total."

Worked example on the slides	Answer

Reading a line graph. Books borrowed: Mon 25, Tue 40, Wed 35, Thu 60, Fri 50. Scale: each line is worth 10. Monday is halfway between 20 and 30. Changes: up 15, down 5, up 25, down 10. Which is the biggest increase?	Wednesday to Thursday: up 25. It is the steepest line going up.
Missing values in a table. Use the row totals first, then add the columns. Check: both ways must give the same grand total.	Ali 35, Mei 32, Ravi 67. June 98, July 80, altogether 178.

SAY *"The key idea today is: scale first, then the changes. Read the scale before you read a point. The steeper the line, the bigger the change. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Reading the scale as 1 for each line when each line is worth 10 (or 5), so 75 is read as 7 and a half.
- Looking only for the biggest increase and missing a bigger decrease, or comparing the heights of points instead of the change between them.
- Comparing two graphs by how steep the lines look without checking where each scale starts and what each line is worth.

30–34 min Brain break

DO Human line graph. The tutor tells a story ("the temperature rises slowly, then drops fast"). Move your arm like the line: slow climb, fast drop, flat.

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Florence Nightingale (Britain (born in Florence, Italy), 1820-1910). Today you read tables and line graphs, and Florence used tables and charts to make numbers easy to understand. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: 9 am to 12 noon: it rose by 5 degrees (31 - 26). Noon to 3 pm: it rose by only 2 degrees. So the biggest rise was 9 am to 12 noon, by 5 degrees. (Between 3 pm and 6 pm it fell by 4 degrees.)

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check

page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Point to a line and ask: "Is this an increase or a decrease? By how much?"
- Ask: "What is each line on the side worth? How do you know?"
- Ask: "How can you check your missing number?"

IF STUCK Hint 1: "Find what one line on the scale is worth before you read anything."

Hint 2: "Write the change between each pair of neighbours, with up or down."

Hint 3: "Add the numbers you know, then take them away from the total."

DO Choosing: after question 2 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask "How did you decide?" and "Is there another way?" rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A table shows that Class 4A has 28 boys and 6 more girls than boys. Class 4B has 63 pupils, with 5 more girls than boys. Which class has more girls?	Neither: both have 34 girls. (4A: 28 + 6 = 34. 4B: 63 – 5 = 58, 58 ÷ 2 = 29 boys, 29 + 5 = 34 girls.)
Workbook Think harder 2: On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?	100 (40 + 20 + 20 + 20)
Workbook Think harder 3: A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?	65 (300 – 65 – 80 – 90)
Workbook Think harder 4: Ali's height: age 7, 120 cm; age 8, 127 cm; age 9, 131 cm; age 10, 137 cm. In which year did he grow the most? How much did he grow altogether from age 7 to age 10?	From age 7 to 8 (7 cm). Altogether 17 cm.

Workbook Go further 1: Four teams scored 100 points altogether. Team A scored twice as many as Team B. Team C scored 10 more than Team B. Team D scored 22. How many points did each team score?	A 34, B 17, C 27, D 22. (A + B + C = 78: four equal parts and 10 more, so one part is 68 ÷ 4 = 17.)
Workbook Go further 2: Tank A has 80 litres of water and loses 6 litres every hour. Tank B has 20 litres and gains 9 litres every hour. On a line graph the two lines cross. After how many hours do the tanks hold the same amount? How much is in each?	After 4 hours: 56 litres each. The gap of 60 litres closes by 15 litres every hour.
Workbook Go further 3: Four numbers: 52, 54, 56, 58. Choose where the scale starts and how much each line is worth, so that a line graph of these numbers looks (a) like a huge increase and (b) like hardly any increase. Write both scales and a sentence a shop owner might say with each graph. Which graph is fair? Why?	Answers vary. (a) Start at 50 with each line worth 1: the line climbs steeply. (b) Start at 0 with each line worth 20: the line is almost flat. A fair graph starts at 0 or says clearly that it does not, so the viewer can see the real change of 6. Both graphs show the same numbers.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Two graphs show the same numbers, but one line looks much steeper. How can that be?"*

"How would you draw a fair graph if you wanted to compare two shops?"

Discussion question	Listen for
Two graphs show the same numbers, but one line looks much steeper. How can that be? How would you draw a fair graph?	The scales are different. If a scale starts at a big number, or each line is worth very little, a small change looks huge. To compare fairly, use the same scale for both graphs, starting at 0.

85–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Visitors: Mon 35, Tue 50, Wed 20, Thu 45. Between which two days was the biggest change? Was it an increase or a decrease?	Tuesday to Wednesday: a decrease of 30.

SAY *"Today you read graphs like detectives: scale first, then the changes, and you caught a graph that was trying to trick you."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

88–90 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I read the scale first. The hardest part was reading between the lines on the scale. What helped me: writing what each line was worth." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Draw two line graphs of the same numbers (52, 54, 56, 58): one that makes the change look huge and one that makes it look tiny. Write a sentence a shop owner might say with each graph.

Answer key (workbook)

Q1 (a) 75 cups; 55 cups (25 + 30) (b) Wednesday to Thursday, up 25 (c) Friday to Saturday, down 40 (d) It is the steepest line going down. A reason such as: the office workers do not come to work on Saturday and Sunday.

Q2 4A Week 2: 47. 4B Week 1: 41. 4C total: 97. Totals: Week 1 124, Week 2 128, altogether 252. (a) 4B (41 is more than 29) (b) 146 (128 + 6 + 6 + 6)

Q3 (a) $40; $10 (b) Week 5 ($50 each); Week 6 (c) $15 (Ali $80, Mei $65)

Q4 Day 1: 10, Day 3: 18, Day 4: 22. (a) 4 more each day (26 − 14 = 12 over 3 days); 90 pages

Q5 (a) Shop A: 60. Shop B: 12. (b) No. The scales are different: Shop A's starts at 0 and each line is 20; Shop B's starts at 40 and each line is 4. Shop A's sales went up by 48 more than Shop B's.

Investigation Sales have doubled! (What if we change ...? What stays the same?) Easy first step: On the Graph grid, write 0, 10, 20 ... up the side. Plot the points (May, 50) and (June, 100) and join them. Model answer: (a) Points (May, 50) and (June, 100); on a scale of 0 to 100 in 10s the line rises through 5 lines, a fair picture. (b) Big: "Sales have doubled!" (100 is twice 50). Small: "Shop sells 50 more cases in June" or "Sales up by 50 cases". Both are TRUE, because 100 is twice 50 and 100 is 50 more than 50. Accept any two true headlines; check each one against the numbers. (c) The most trustworthy report gives both

numbers (50 and 100) and the real rise, and shows a graph that starts at 0 with equal steps and the scale written on it. Look for a child who checks that a headline is TRUE before saying how it sounds, and who notices that the numbers did not change, only the words did. Easy first step: plot the two points on a 0 to 100 scale in 10s.

Go further Draw the same graph again with a different scale. Does the story look different? Why? Yes. A scale with small steps, or one that starts above 0, makes the line look steeper; big steps make it look flatter. The numbers are the same: only the picture changed.

Story corner try-it Florence Nightingale: Temperatures: 9 am 26 degrees, noon 31 degrees, 3 pm 33 degrees, 6 pm 29 degrees. Between which two times did it rise the most, and by how much? Answer: 9 am to 12 noon: it rose by 5 degrees (31 - 26). Noon to 3 pm: it rose by only 2 degrees. So the biggest rise was 9 am to 12 noon, by 5 degrees. (Between 3 pm and 6 pm it fell by 4 degrees.)

Sneak-peek puzzles (1) Tuesday sold the most (45). Wednesday sold the fewest (25). (2) 125. Halfway between 100 and 150 is 100 + 25.

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed heuristics • Workbook: Lesson 10

Objective	Choose a way into a non-routine problem (guess and check with a table, suppose they are all the same, work backwards, a logic grid, draw it) and use it on competition-style puzzles, checking every answer against every fact.
Thinking skills	Choosing a strategy, guess and check, supposition, working backwards, logical deduction, checking and explaining
Materials	Four puzzle cards (A to D) for the explore stations (pairs choose three), mini whiteboards, counters or cubes, certificates, stamps.
Online set-up	Pairs go to breakout rooms with the puzzle cards (each pair chooses three of the four) and volunteers report back; the tutor ticks and crosses the logic grid on the slide as children call out.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome to Challenge Day! Today there is no new topic. Instead, you get the puzzles that mathematicians love, the ones where you do not know how to start."*

"Good problem solvers are not people who know every answer. They know how to START. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. Keep it to one minute.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"Two sneak-peek puzzles about ways in to a puzzle. These are ideas we have not learnt yet, so guessing and "I do not know" are welcome. Nobody is marked. Write your answer on your own paper and hold it up to the camera when I say."*

DO Show the Sneak-peek puzzles slide. Give about 90 seconds per question. Question 1 uses objects, a drawing or counting, so every child can start. Question 2 is a thinking question. Do not teach and do not say who is right yet.

Answers are shown only to you: each child writes on their own paper and holds it to the camera at the same moment (or shows only you, on camera). Keep every part of this in the shared session, and never ask for boards up in front of the group. Walk the group through the answers at the end, using the slide, naming no child, so everyone hears the reasoning.

Use what you see only to plan your help. It does not decide who works on which page. Everyone starts on Start here (page 1) and the invitation is for the WHOLE group: "After question 3 you may choose: carry on, or try Think harder. When you have finished Start here you can choose Think

harder, Go further or the Investigation." Never single out or name a child, never announce who is on which page, and do not split a pair unless both children agree. If one child of a pair moves on and the other does not, join the other child to another pair or work with them yourself straight away. Children who choose Think harder or Go further stay in the same room and still join the checks at minutes 50 and 76, talk time, the exit ticket and the reflection.

Sneak-peek puzzle	Answer
Sneak-peek 1 (objects, drawing or counting): Draw 5 ducks. A duck has 2 legs. How many legs are there altogether? Now change ONE duck into a rabbit (a rabbit has 4 legs). How many legs now?	10 legs, then 12 legs. Changing a duck into a rabbit adds 2 legs.
Sneak-peek 2 (thinking): A garden has only ducks (2 legs) and rabbits (4 legs). There are 8 animals and 22 legs. How many rabbits are there?	3 rabbits (and 5 ducks): 5 × 2 + 3 × 4 = 22. Suppose all 8 are ducks: 16 legs, 6 too few; each rabbit instead of a duck adds 2 legs, so 3 rabbits.

DO A good reason sounds like: Pick a way in: guess and check with a table, suppose they are all the same, or work backwards from the end. Then check every fact.

4–8 min Warm-up game and quick questions

DO Who am I? Four clues, one at a time, on a 1 to 100 grid. Cross out every number that does not fit each clue until one is left. (Today play one short round, about 2 minutes, because the Sneak-peek puzzles came first.)

SAY *"I am thinking of a number from 1 to 100. I will give you one clue at a time. After each clue, cross out every number it cannot be."*

"Clue 1: I am a 2-digit number. Which numbers can you cross out straight away?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Buses every 6 min and every 8 min leave together at 9:00. Next together?	9:24
Round 45 678 to the nearest thousand.	46 000
3/4 of 24 = ?	18
0.6 × 4 = ?	2.4
Turn clockwise from N to SW. How many degrees?	225°

8–18 min Explore (in pairs, with objects)

SAY *"Look at the four puzzle cards. Which one do you think will be hardest? Write the letter in the corner of your whiteboard. We will check at the end."*

DO 1. Four puzzle cards, A to D. Start with any card.

2. Read the card aloud with your partner.

3. Try something: a guess, a drawing or a table.

4. Write the answer AND how you got it.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK try 10 × 11, then a bigger pair. Is the answer too big or too small?

what if all 10 bugs were beetles? How many legs would that be?

start at 43 and undo the last thing first.

Finished early: make up a new card for one of the stations.

DO At about minute 16, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How did you start card B?	Suppose all 10 are beetles: 60 legs, 8 short; each spider adds 2 legs, so 4 spiders and 6 beetles.
Which card was hardest? What helped?	Open answers: a table, a drawing, or starting at the end.
Could any card have two answers? How do you know?	No: each one was checked. A: 14 × 15 = 210. C: 18. D: Ali fish, Mei cat, Ravi dog.

18–30 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Two next-door numbers multiply to 342. I estimate: 20 × 20 is 400, too big. Try 17 × 18 = 306, too small. 18 × 19 = 342. My table tells me which way to go after each try."*

"15 bicycles and tricycles with 38 wheels. Suppose they are ALL bicycles: 30 wheels. I am 8 wheels short. Each time I swap a bicycle for a tricycle I get 1 more wheel. So I swap 8: 8 tricycles and 7 bicycles. Check: 14 + 24 = 38."

"Siti spent half, then $4, and had $11. I start at the end and undo each step, last one first: add the 4 back, $15; double it, $30. Check forwards: half of 30 is 15, take 4 is 11."

Worked example on the slides	Answer
Guess and check with a table. Two next-door numbers multiply to make 342. Each try tells you to go up or down.	18 and 19

Suppose they are all the same. 15 bicycles and tricycles have 38 wheels. Suppose all are bicycles: 30 wheels. 8 short. Each swap to a tricycle adds 1 wheel.	8 tricycles and 7 bicycles. Check: 14 + 24 = 38.

SAY *"The key idea today is: pick a way in. Guess and check with a table, suppose they are all the same, or work backwards from the end. Say it with me."*

DO Show the key idea slide at minutes 29 to 30. The "your turn" slides are optional: skip them if you are short of time. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Guessing at random instead of using a table, so each try does not tell them whether to go up or down.
- Working backwards but doing the same operation again (taking away 4 instead of adding 4) or undoing the steps in the wrong order.
- Queues: adding the two positions (5th and 8th makes 13) and counting the person twice; logic puzzles: stopping before every clue has been used.

30–34 min Brain break

DO Heads and legs! The tutor calls "3 chickens and 2 goats": stamp that many legs (14). Then "5 animals, 16 legs": who can say the animals?

Water. Toilet break for anyone who needs it. Keep it to four minutes (or two minutes of movement and then the optional story below).

DO Story break (optional, 2 minutes): today's mathematician is Terence Tao (Australia (born in Adelaide); now also works in the USA, born 1975). Today is Challenge Day, with competition-style puzzles like the ones in the Olympiad that Terence Tao entered three years in a row. Show the Story break slide (it follows the Brain break slide), read the story aloud or ask a volunteer, then offer the Try it question to the whole group. Nobody is marked or ranked.

To skip: say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on the story. Try it answer: 6 hens and 4 cows. Check: 6 + 4 = 10 heads; 6 x 2 + 4 x 4 = 12 + 16 = 28 legs. (Try 5 cows: 20 + 10 = 30 legs, too many; 3 cows: 12 + 14 = 26, too few.)

34–80 min Worksheet (pairs, then alone)

DO Minute 34: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Minutes 34 to 50: pairs work together on the first page (online, allow about 2 minutes to move into and out of breakout rooms, or run page 1 as a whole-group screen-share). Minutes 50 to 53: check page 1 together. Minutes 53 to 58: the short teaching before page 2. Minutes 58 to 76: each child works alone on the second page. Minutes 76 to 80: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “What is your way in? Why did you choose it?”
- Ask: “Does your answer fit EVERY fact in the question? Show me.”
- Ask: “Could there be another answer? How do you know?”

IF STUCK Hint 1: “Make a table and try a number. Was it too big or too small?”

Hint 2: “What if they were all the same? How far off are you?”

Hint 3: “Start at the end and undo, last step first.”

DO Choosing: after question 3 on page 1 every child may carry on or try Think harder. Make the offer to the whole group, never to one child by name. The Think harder page, the Go further page and the Investigation have no fixed time slot: any child may choose them when they have finished Start here, and they are good for take-home. Page 2 is on your own and has a printed note about which questions to do first, so nobody has to rush. Every lesson has an Investigation on page 3, with an easy first step and a model answer in the answer key below: it has several valid answers, so ask “How did you decide?” and “Is there another way?” rather than marking it right or wrong.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?	7 tricycles (and 13 bicycles). All bicycles: 40 wheels, 7 short, each swap adds 1.
Workbook Think harder 2: Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?	24 and 26
Workbook Think harder 3: Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?	$51 (18 × 2 = 36, 36 + 15 = 51)
Workbook Think harder 4: Ali, Ben and Siti are 9, 10 and 11 years old. Ben is older than Siti. Siti is not the youngest. How old is each child?	Ben 11, Siti 10, Ali 9.
Workbook Go further 1: A hawker centre has 15 tables. Some seat 4 people and some seat 6 people. There are 76 seats altogether. How many tables seat 6?	8 tables seat 6 (and 7 seat 4). All seat 4: 60 seats, 16 short, each swap adds 2.
Workbook Go further 2: Mei gave Ali half of her stickers and 2 more. Then she gave Ben half of the stickers left and 2 more. She had 5 stickers left. How many did she have at first?	32. Work backwards: (5 + 2) × 2 = 14, (14 + 2) × 2 = 32.

Workbook Go further 3: Make up your own heads-and-legs puzzle (chickens and goats, or bicycles and cars) that has exactly ONE answer. Check it by trying every number of one animal. Then explain why a farmer who counts 12 heads and 35 legs on a farm with only chickens and goats must have counted wrong.	Puzzles will vary: check by trying every possible number. Model: 10 vehicles, 32 wheels, cars (4 wheels) and motorcycles (2 wheels): 6 cars and 4 motorcycles is the only way. Why 35 is impossible: every chicken has 2 legs and every goat has 4, so the total number of legs is always even, and 35 is odd.

These are the questions on the Think harder page and the Go further page in the workbook. More are in the separate Question Bank.

80–85 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first. No cold-calling, nobody has to speak, and passing is always fine. Never keep a tally or scoreboard in front of the group. Praise the reasons, not the speed.

SAY *"Which way in did you use most today?"*

"If you would like to, pick one puzzle and tell us your way in, in two sentences. Volunteers first. You can pass."

Discussion question	Listen for
Which way in did you use most today? If you would like to, pick one puzzle and tell the group your way in, in two sentences. You can pass.	Open answers. Listen for the name of the strategy (guess and check, suppose, work backwards, logic grid, draw it) and a check against every fact.

85–87 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A farm has ducks and goats: 12 heads and 34 legs. How many goats? Show your way in.	5 goats (and 7 ducks). Suppose all ducks: 24 legs, 10 short, each swap adds 2 legs, so 5 swaps. Check: $7 \times 2 + 5 \times 4 = 34$.

SAY *"You are real inquirers now: you can start a problem you have never seen before."*

"Give each child a certificate and their stamped inquiry journal. Tell each one a specific thing they did well this holiday."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

87–89 min Looking back (2 minutes)

SAY *"Turn to page 3, to the small box called Looking back. You can write, tick or pass: you choose. Finish what you like: Something I did well, The hardest part was, What helped me. Two minutes, no talking."*

"Model one yourself first: "Something I did well: I tried a table when I was stuck. The hardest part was choosing which way in to use. What helped me: trying a table first, then checking every fact." Ticking "I am still stuck" is a good thing to do: it tells me where to help."

DO Run it in two minutes. It is never collected, never graded, never corrected and never shown to the group. You may look at a child's box only if the child holds it to your camera. Do not ask anyone to read theirs aloud; if a child wants to share, take one volunteer and thank them for the strategy. If "I am still stuck" is ticked, check in during the open session: a quick visit to the shared breakout room or a hand signal, with a parent nearby. Praise the strategy ("you drew it", "you asked for help"), not the speed or the score.

Extension: for any child who wants more

Write your own heads-and-legs puzzle (for example, bicycles and cars) with exactly one answer. Swap with a partner and solve it by supposing.

Answer key (workbook)

Q1 (a) 23 and 24 (23 × 24 = 552) (b) 2 ways: 9 pens and 2 notebooks ($36 + $14), or 2 pens and 6 notebooks ($8 + $42). Money left for pens: $43, $36, $29, $22, $15, $8, $1; only $36 and $8 are in the 4 times table.

Q2 (a) 36 legs; 14 legs short (b) 2 legs; 7 goats and 11 chickens (check: 22 + 28 = 50) (c) 11 cars (and 13 motorcycles: 44 + 26 = 70)

Q3 (a) 52 (30 + 5 = 35, 35 − 9 = 26, 26 × 2 = 52) (b) 20 (21 × 2 = 42, 42 + 18 = 60, 60 ÷ 3 = 20)

Q4 Ali: nasi lemak. Ben: laksa. Mei: prata. Siti: chicken rice.

Q5 (a) 18 people (6 + 1 + 11) (b) 15th from the back (14 people behind him, 20 − 6 = 14)

Q6 Ben is 10 and his father is 40. (In 5 years the total is 60; 4 equal bars make 60, so Ben will be 15.)

Investigation Ring toss scores (Find all the ways.) Easy first step: List the scores you can get with ONE ring: 3 or 5. Then with TWO rings: 3 + 3, 3 + 5 and 5 + 5. Write each total. Model answer: (a) You cannot get 1, 2, 4 or 7. You can get 3, 5, 6 (3 + 3), 8 (3 + 5), 9 (3 + 3 + 3), 10 (5 + 5) and every score from 8 up to 20 (checked by brute force up to 30). (b) After 7 you can get every score: 8, 9 and 10 can be made, and adding a 3 gives the next ones. (c) With 4 and 6 every score is even, so all the odd scores are impossible, and 2 as well (every even score from 4 up is possible). Why: both rings score even numbers, and even + even is even. Look for a systematic list or table, and for children who can say WHY about the even scores.

Go further What if 10 bugs have 65 legs? Could that happen? Why or why not? No. Spiders have 8 legs and beetles 6: both are even numbers, so the total number of legs is always even, and 65 is odd.

Story corner try-it Terence Tao: Use guess and check: a farm has hens and cows. Together they have 10 heads and 28 legs. How many hens and how many cows? Answer: 6 hens and 4 cows. Check: 6 + 4 =

10 heads; 6 x 2 + 4 x 4 = 12 + 16 = 28 legs. (Try 5 cows: 20 + 10 = 30 legs, too many; 3 cows: 12 + 14 = 26, too few.)

Sneak-peek puzzles (1) 10 legs, then 12 legs. Changing a duck into a rabbit adds 2 legs. (2) 3 rabbits (and 5 ducks): 5 × 2 + 3 × 4 = 22. Suppose all 8 are ducks: 16 legs, 6 too few; each rabbit instead of a duck adds 2 legs, so 3 rabbits.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. Invitations to children go to the whole group and check-ins happen in the open session.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for children whose school year covers Primary 4 topics

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Factor Hunters: factors and multiples
- Lesson 2, Big Number Puzzles: numbers to 100 000
- Lesson 3, Operation Detectives: four operations
- Lesson 4, Fraction Frontier: fractions
- Lesson 5, Decimal Detectives: decimals
- Lesson 6, Angle Turns: angles
- Lesson 7, Area Architects II: area and perimeter
- Lesson 8, Symmetry and Nets: symmetry and nets
- Lesson 9, Data Detectives: tables and line graphs
- Lesson 10, Challenge Day: mixed heuristics

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep scissors, a ruler and a pencil ready.

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It builds problem-solving and thinking skills and is not affiliated with, run by or endorsed by the Ministry of Education or any school.